New technologies in IoT

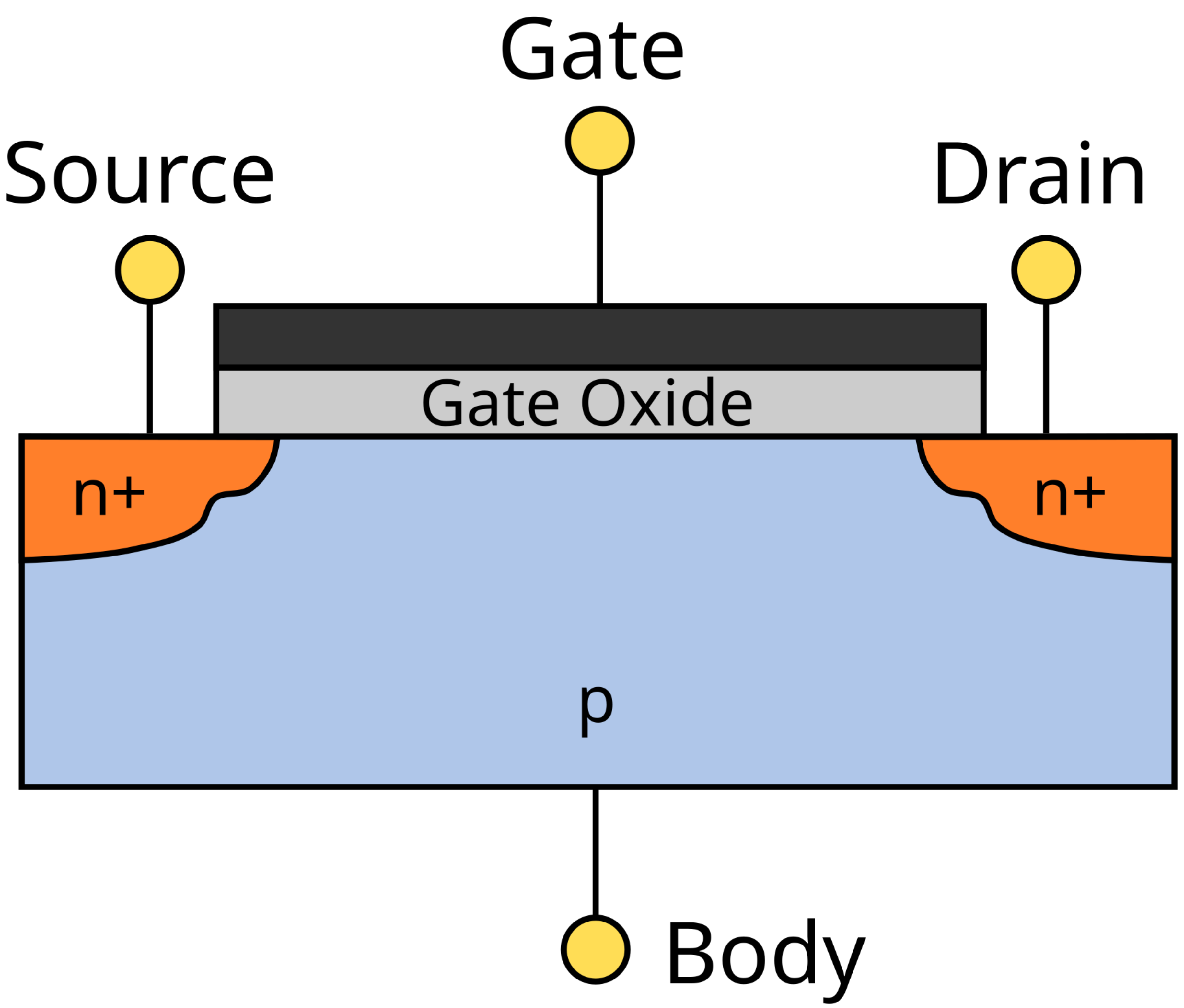
Internet of Things

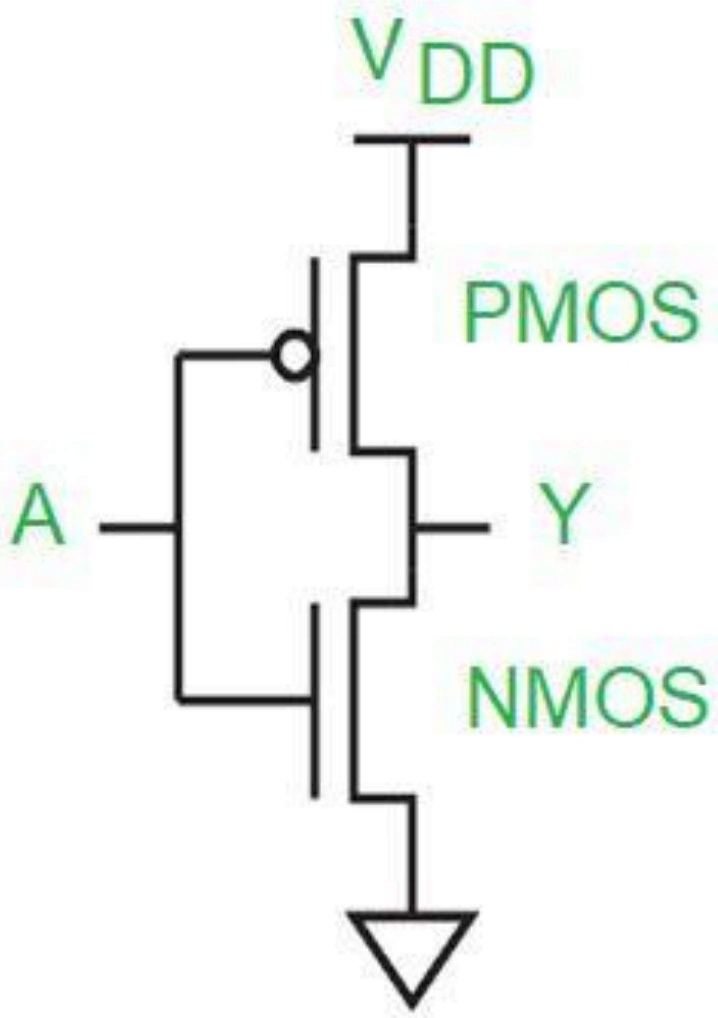
CS5055 - 2025I

PROF.: CWILLAMS@UTEC.EDU.PE

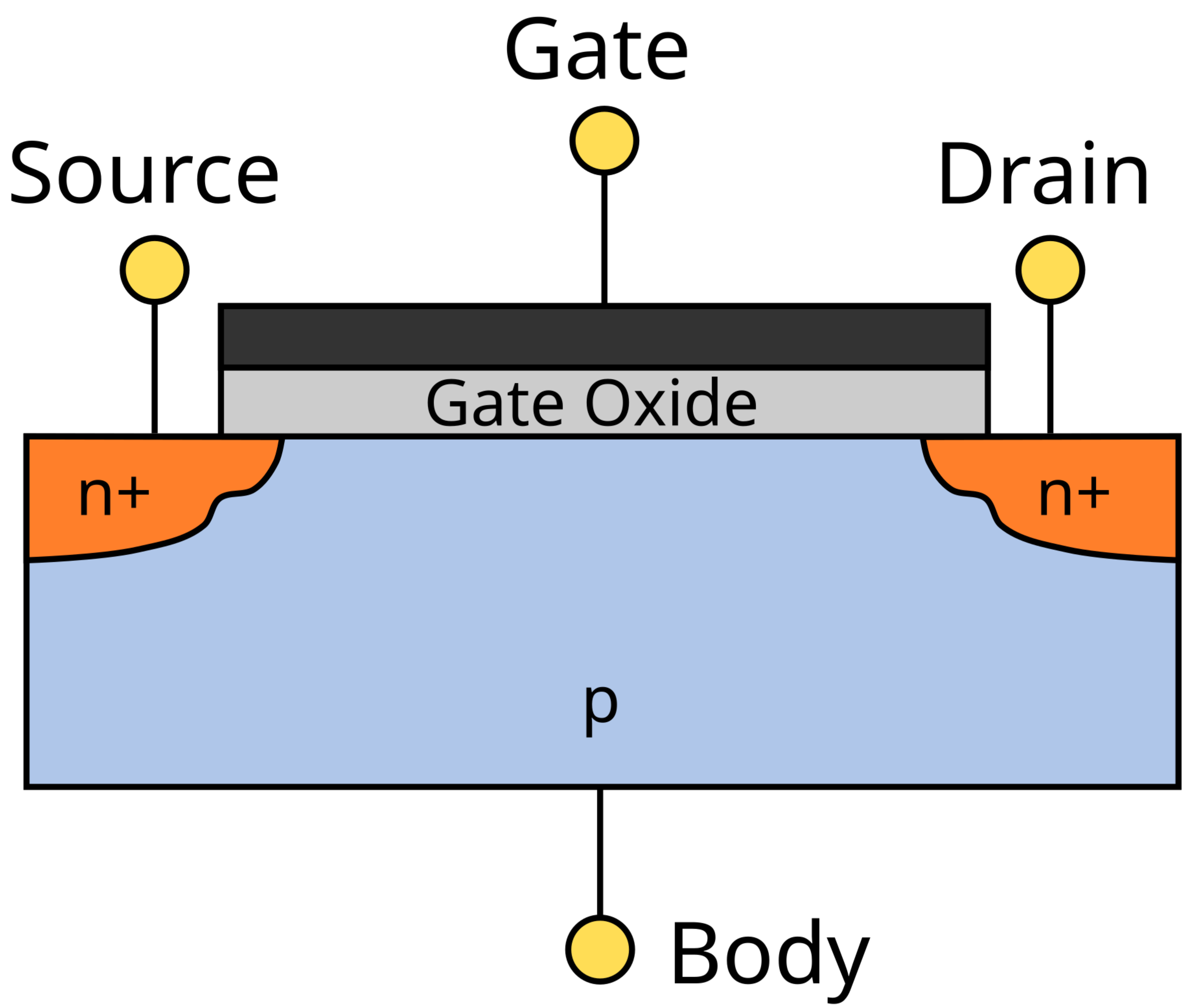
Outline

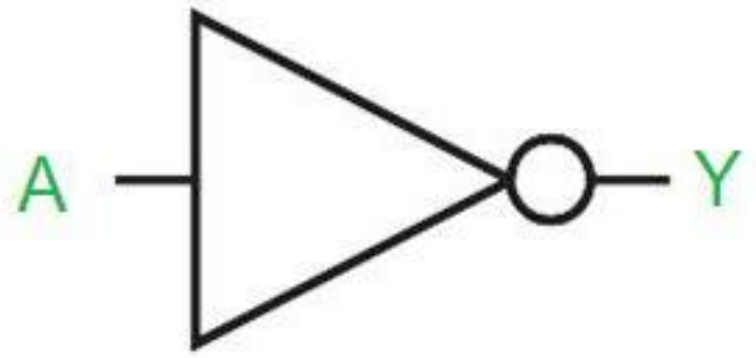
Transistor





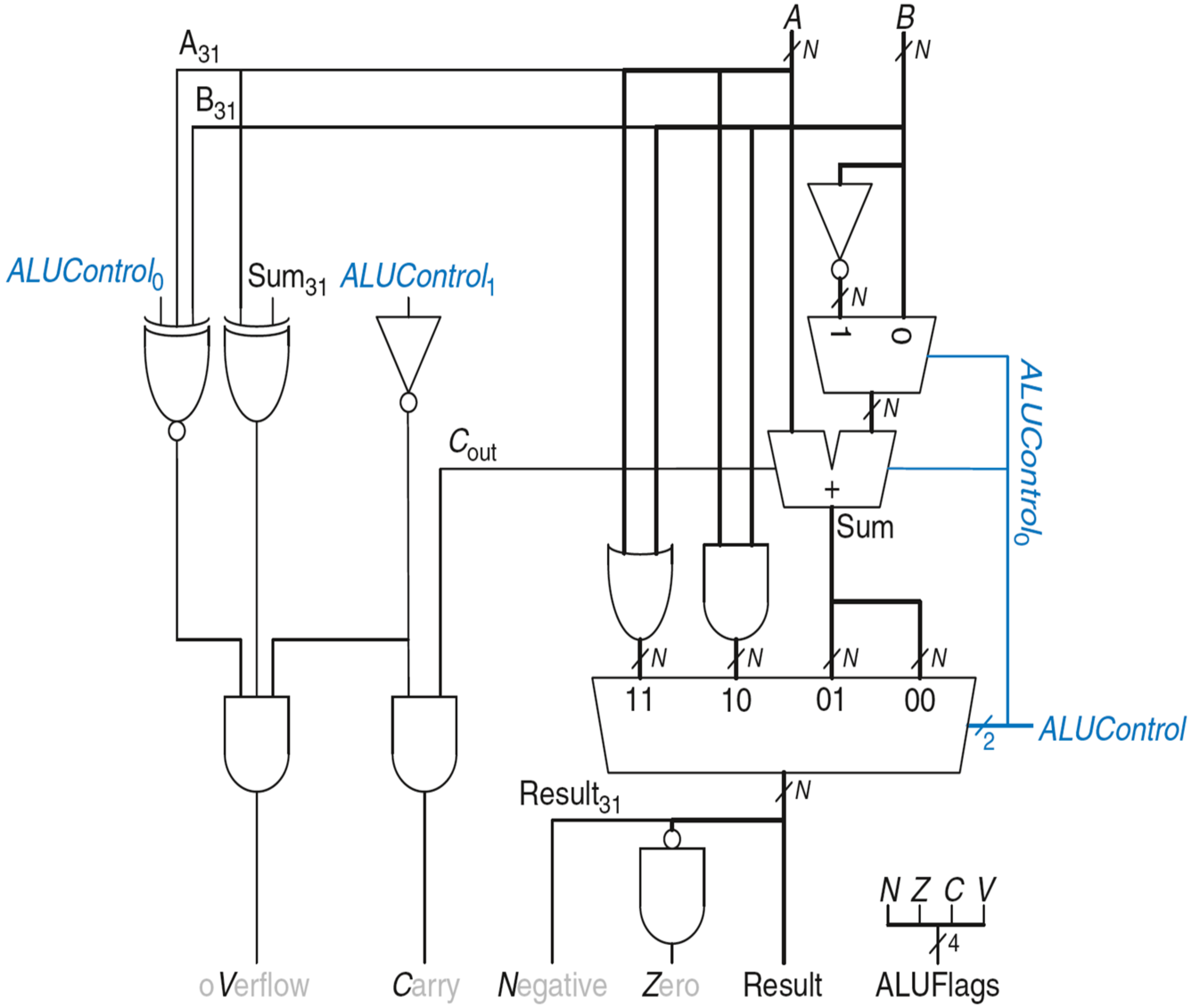
Transistor

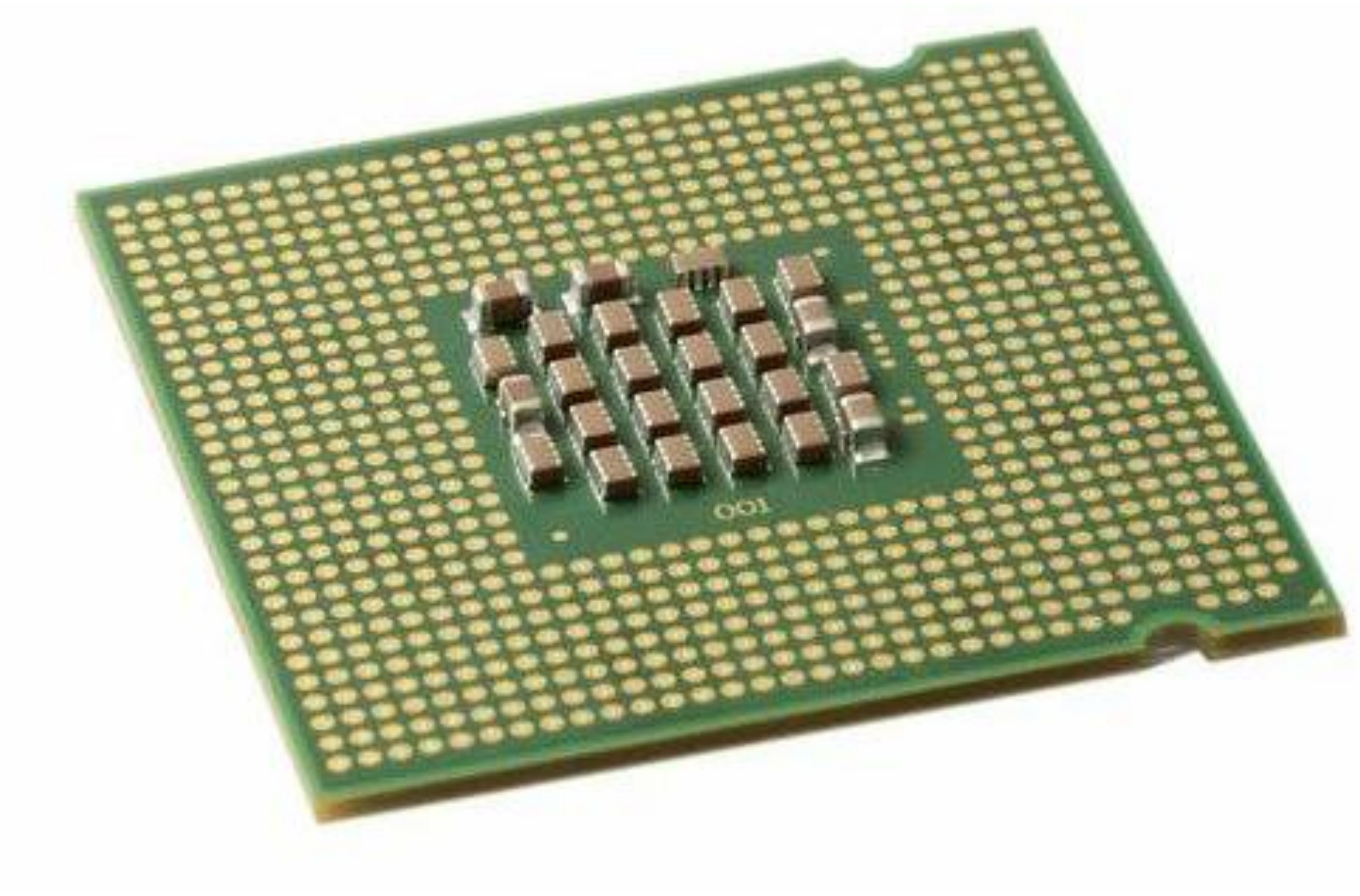




A	Y
0	1
1	0

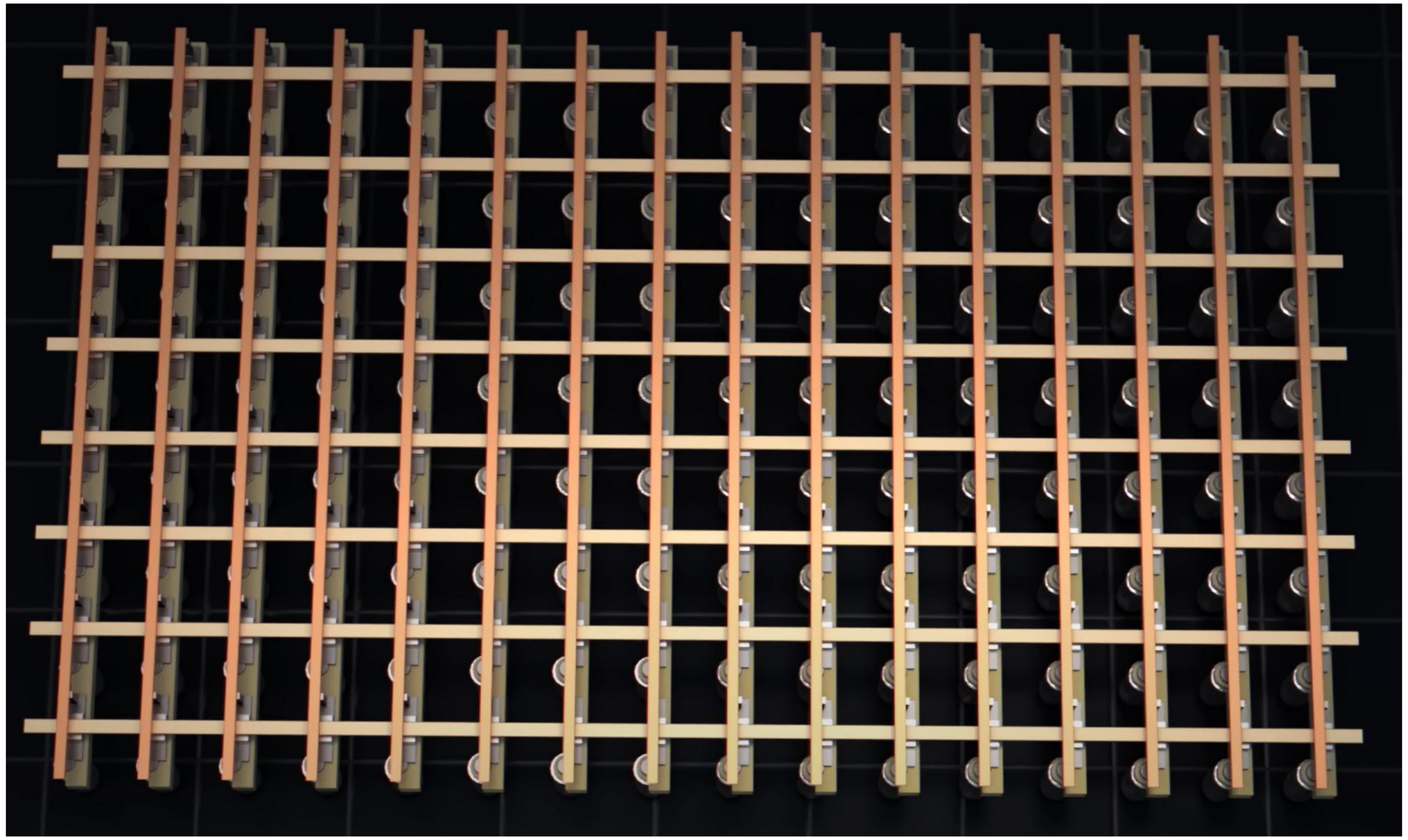
Logic gates - CPU

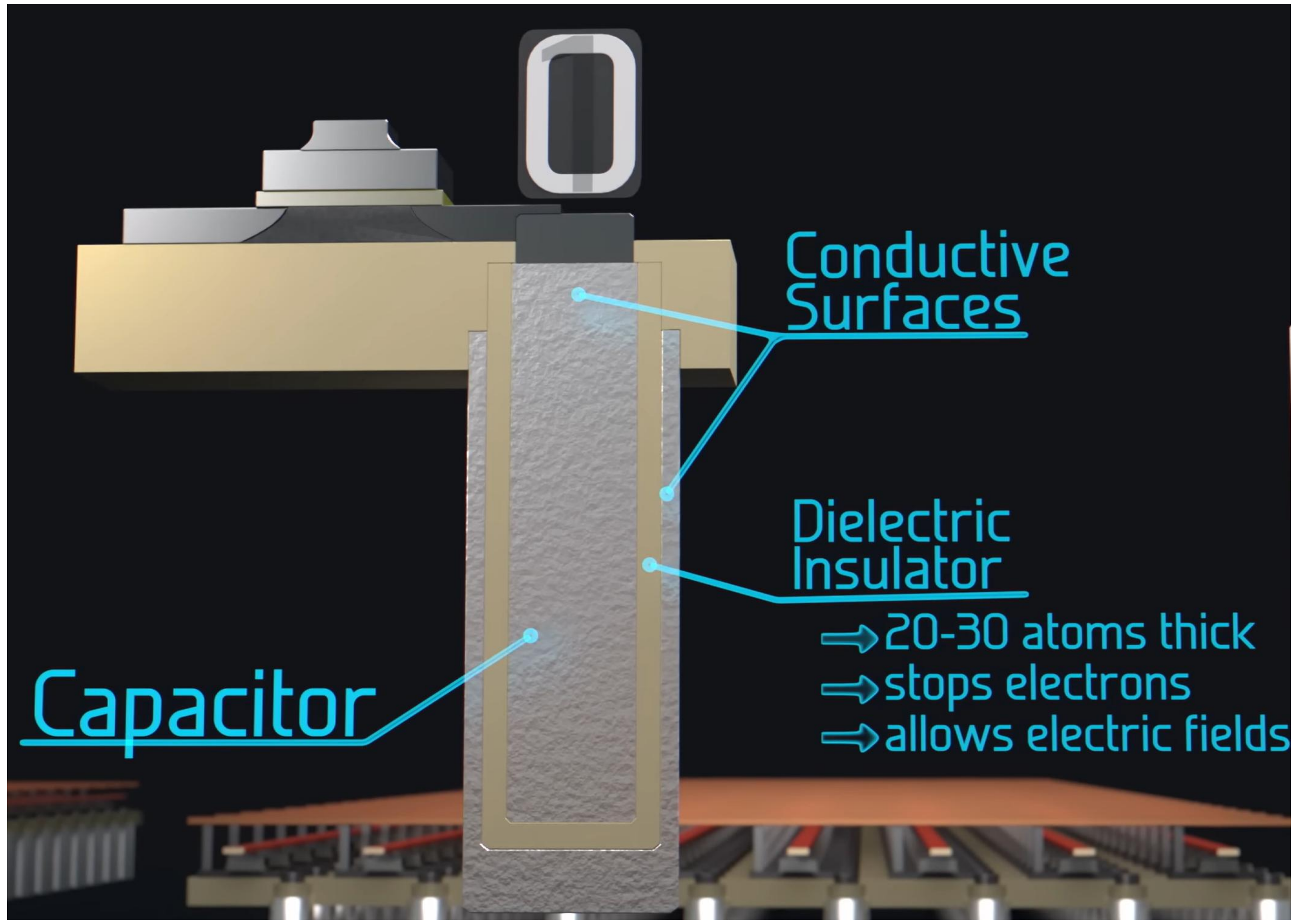




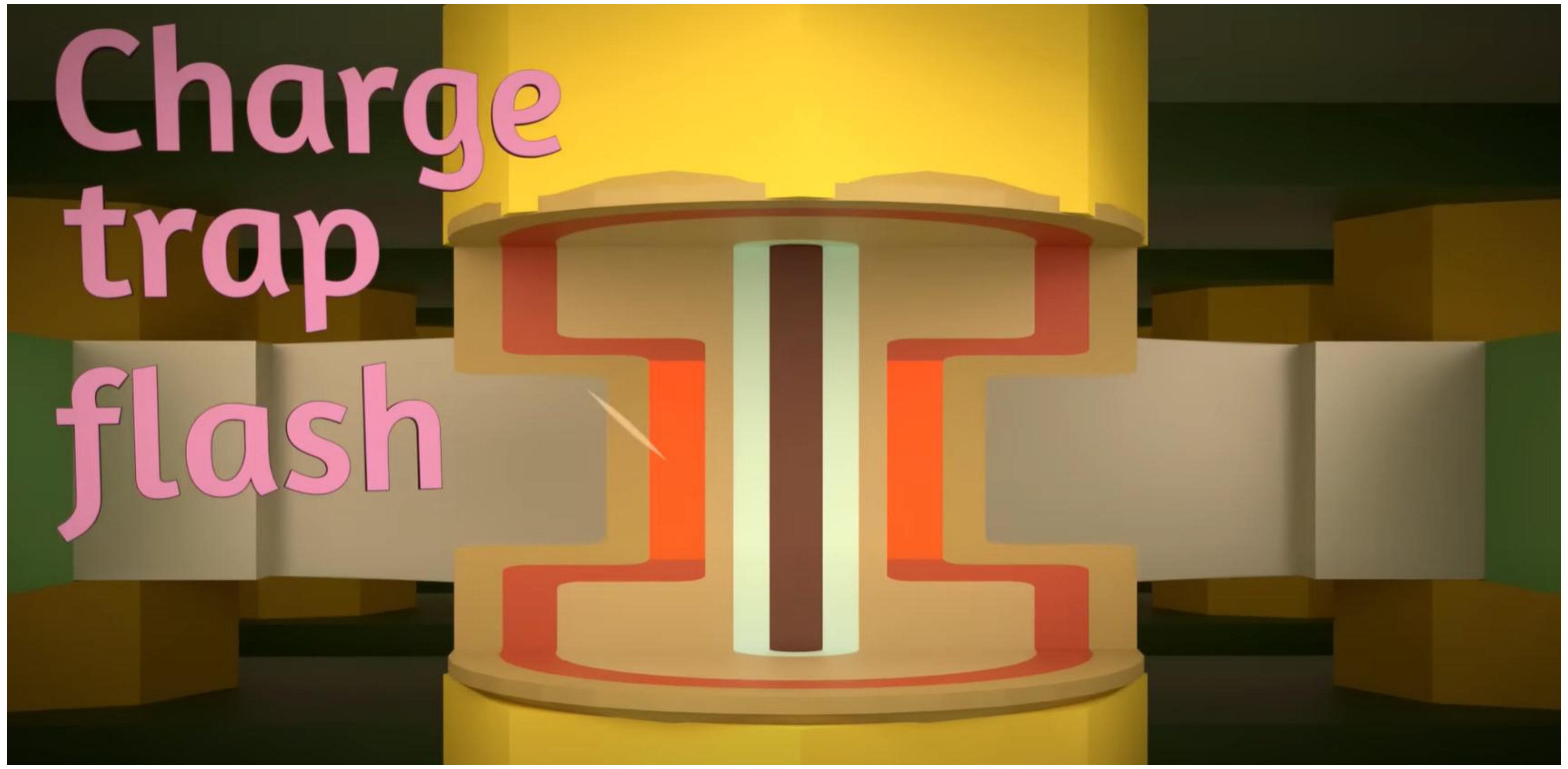
Memory?

RAM - capacitor

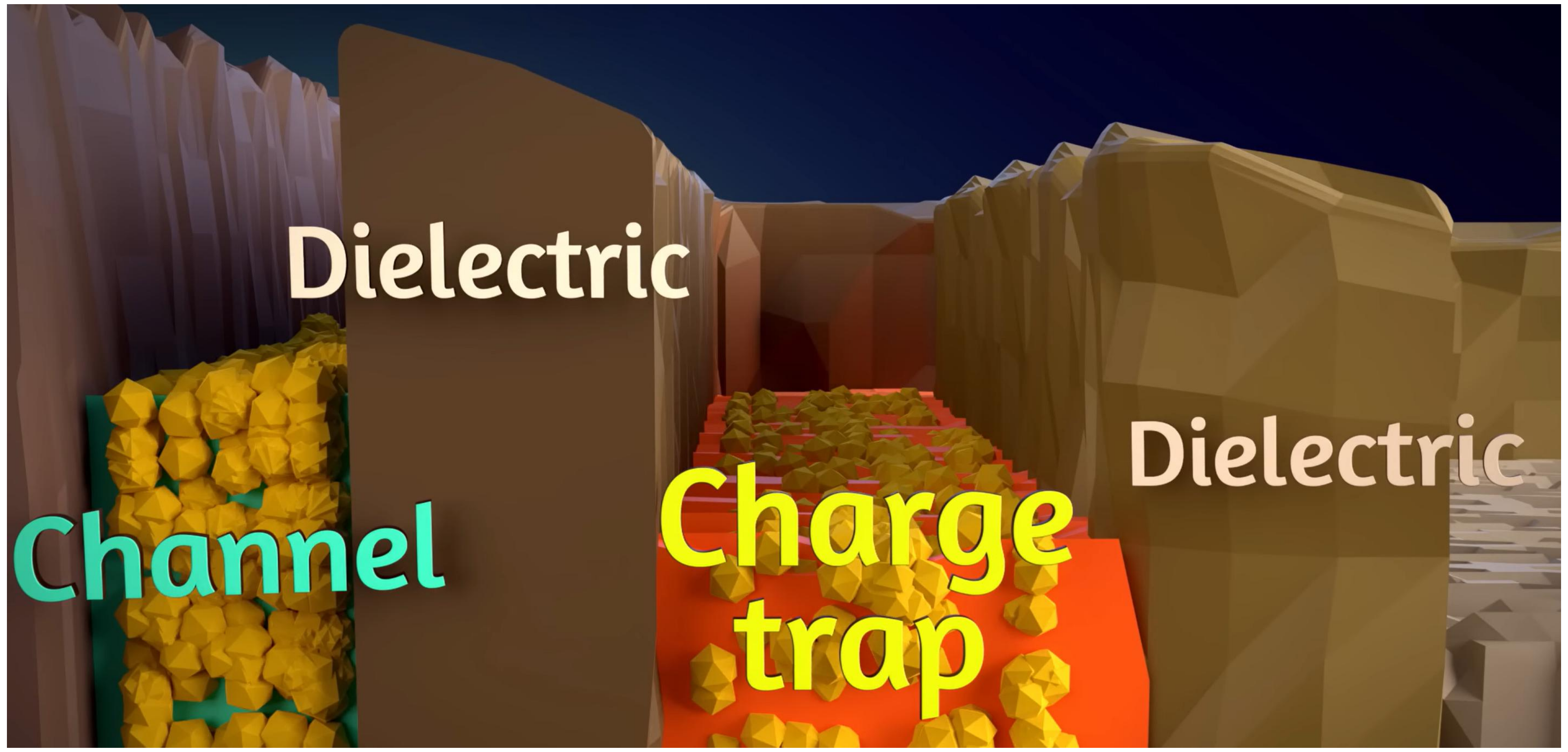




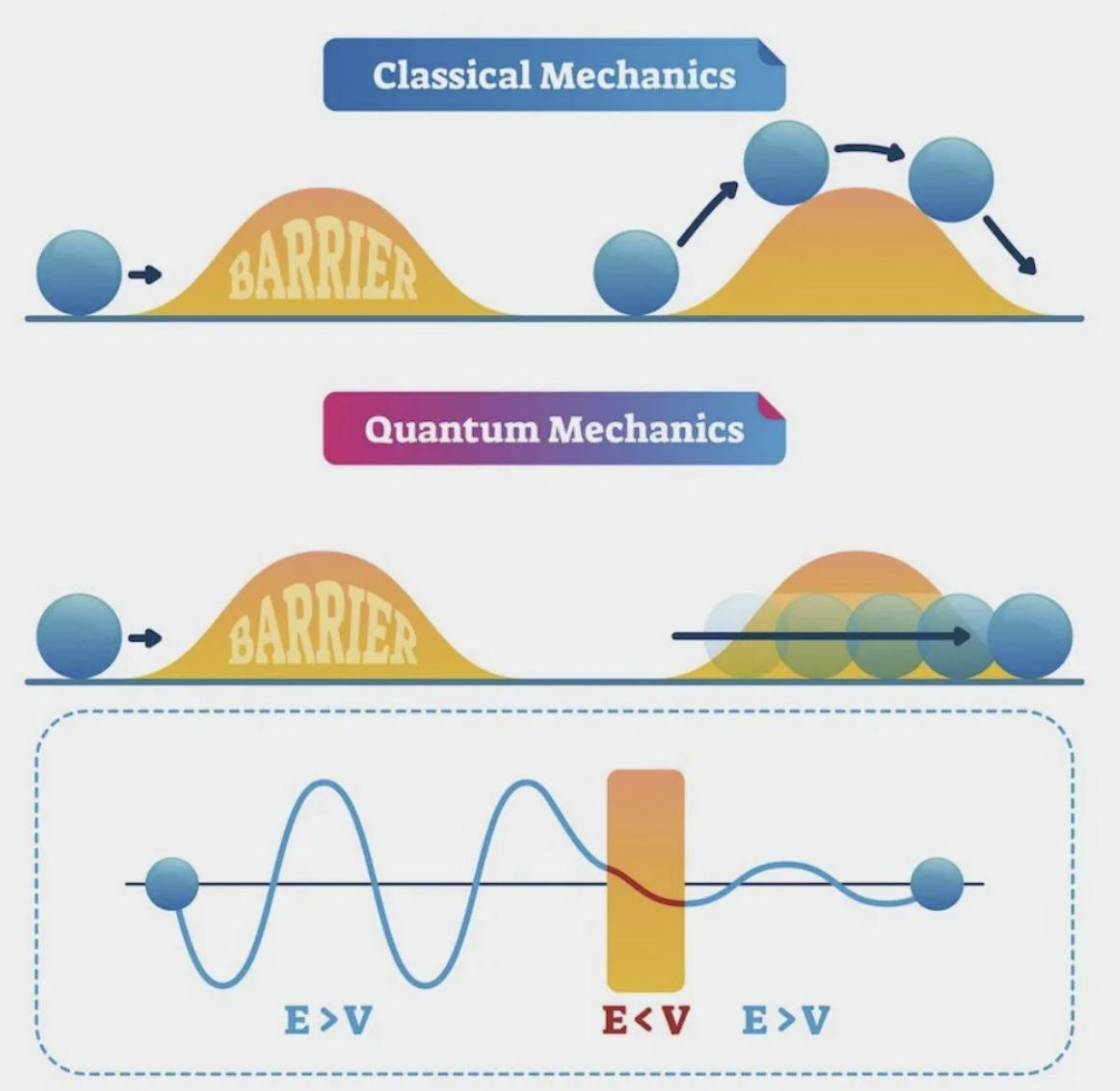
SSD Memory – Charge trap



How does charge cross the dielectric barrier?



Quantum tunneling

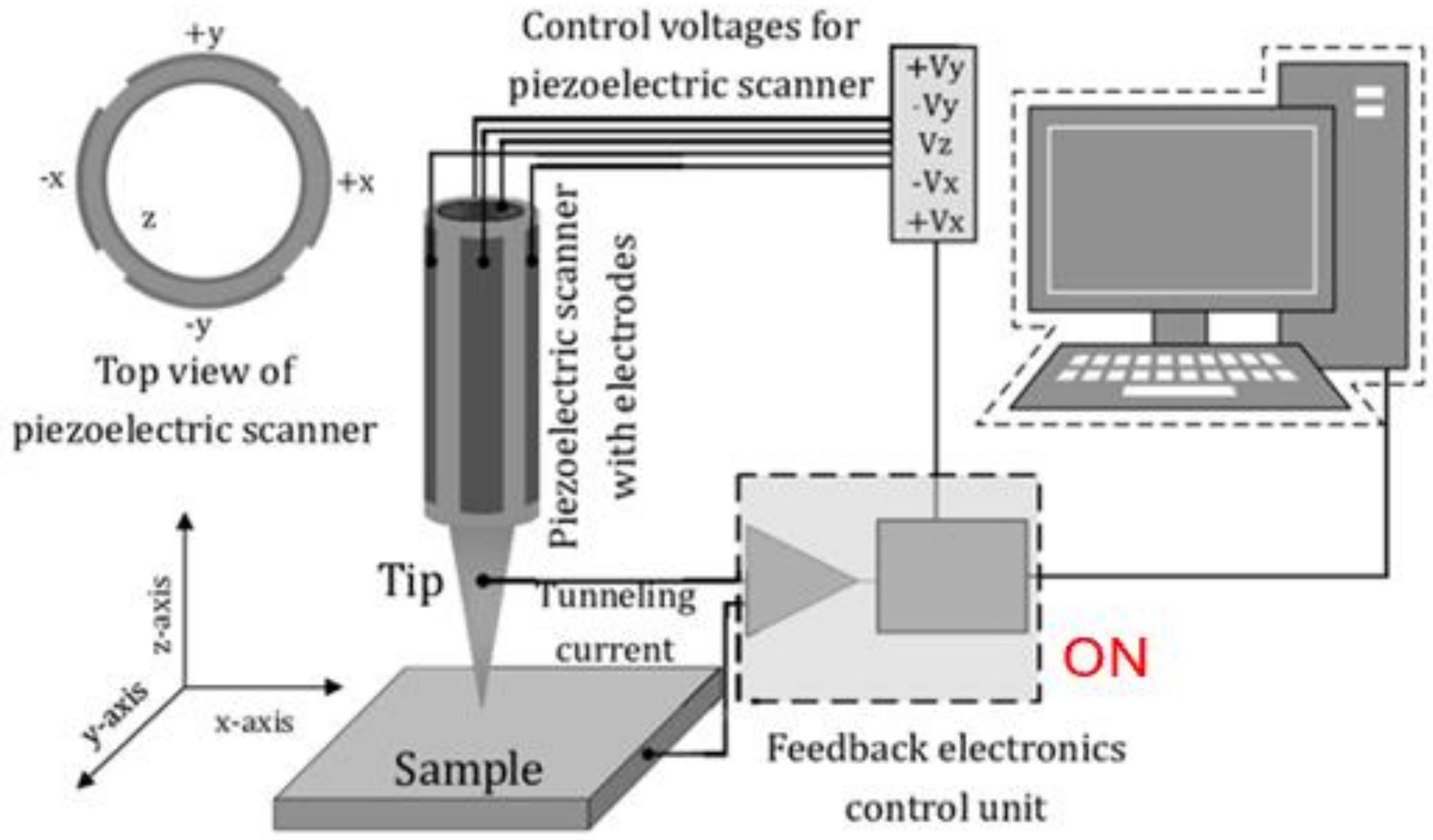


$$I = \frac{2\pi e^2}{\hbar} V_{bias}\ \rho_{\text{tip}}(E_F)\ \rho_{\text{sam}}(E_F)\ \overline{M^2}(E_F)$$

$$I \propto \rho(z) \propto e^{-2\kappa z}$$

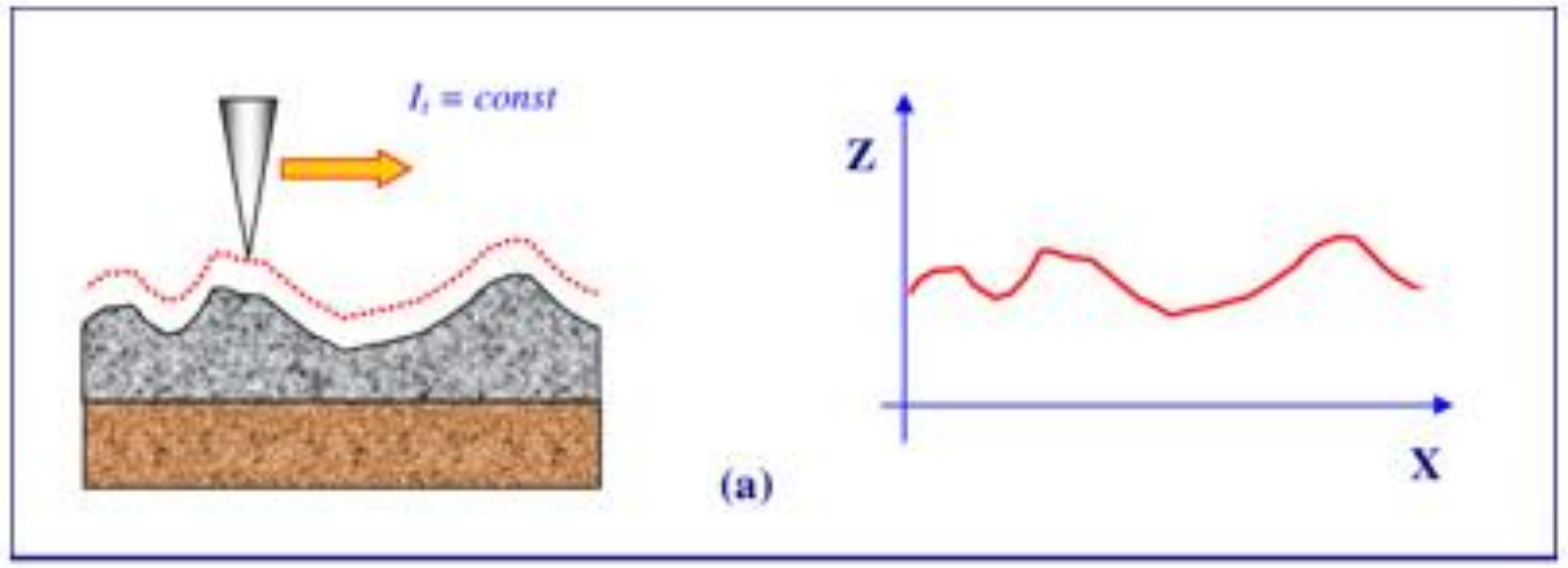
Scanning Tunneling Microscope

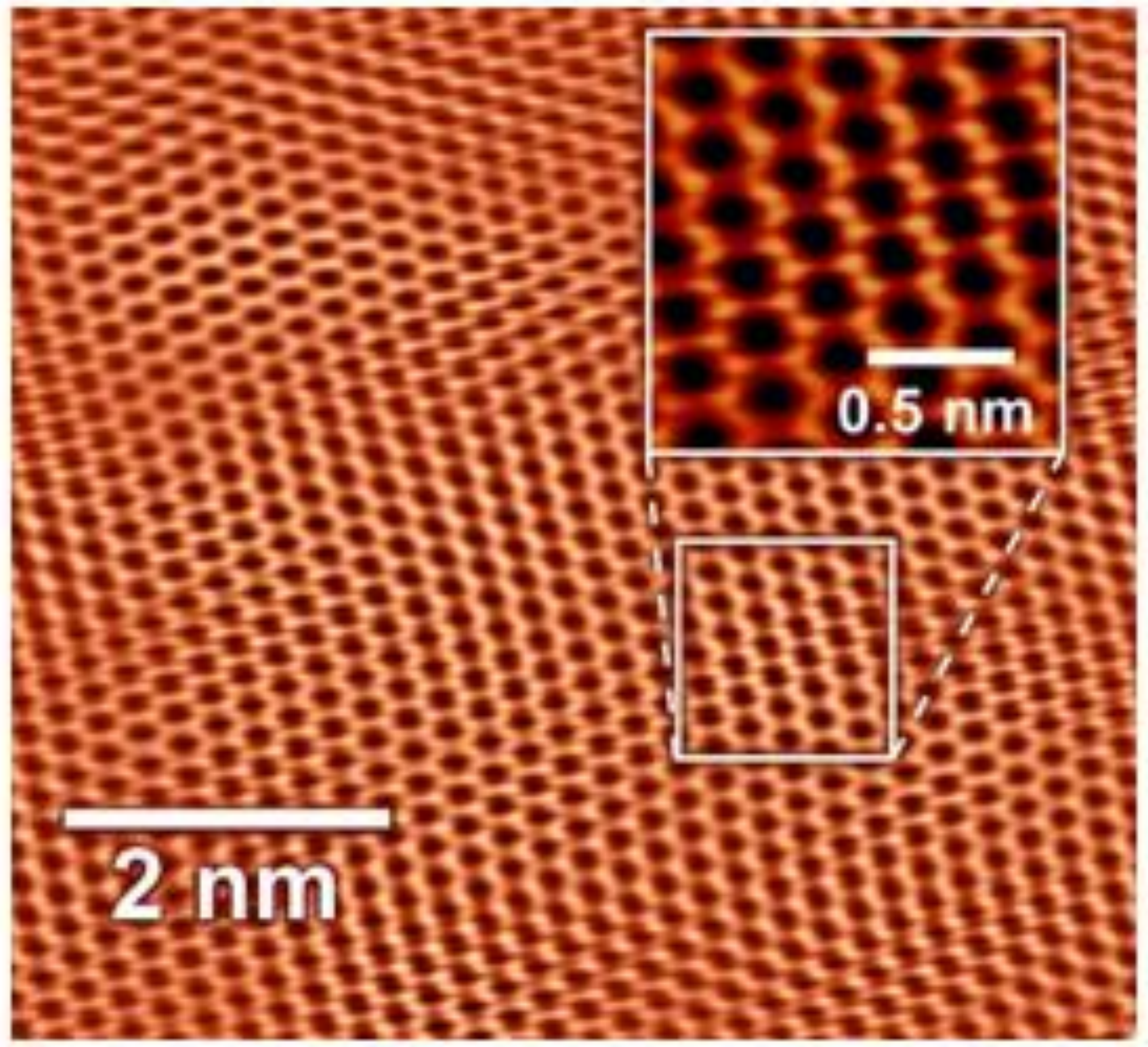
$$I \propto \rho(z) \propto e^{-2\kappa z}$$



Schematic drawing of a STM set-up.

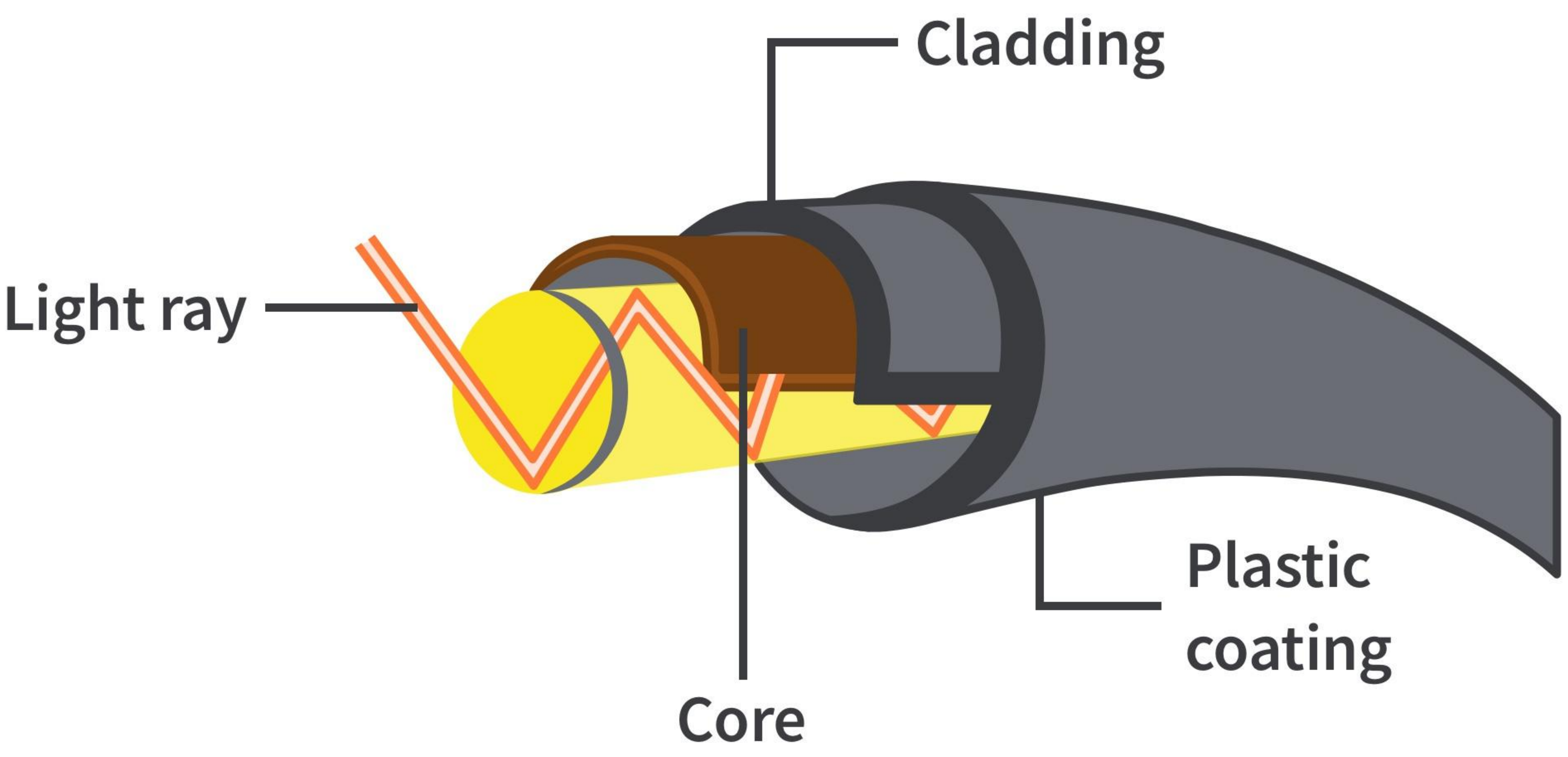
Mironov, Victor. (2014). Fundamentals of Scanning Probe Microscopy.





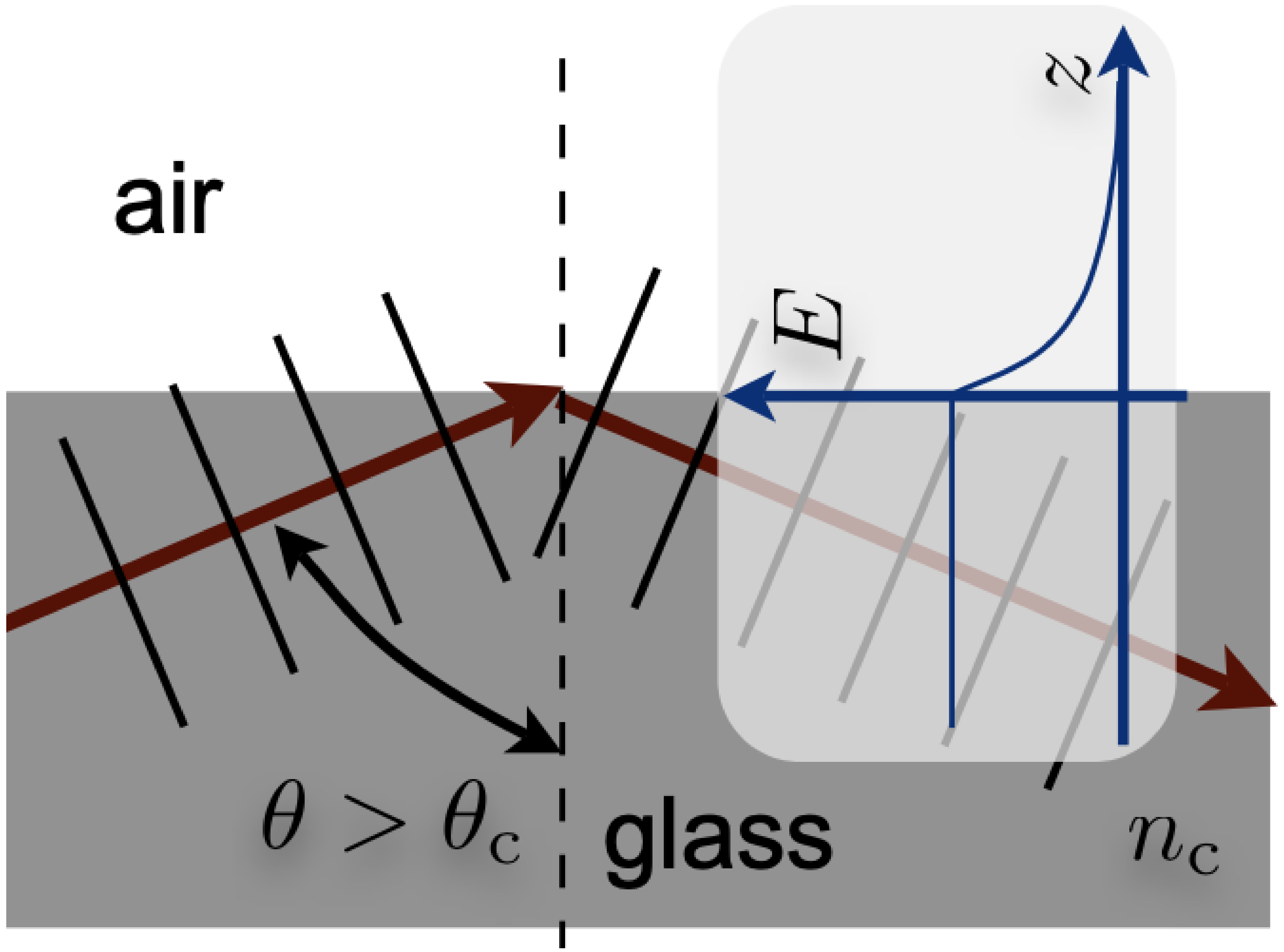
Outline

Fibra óptica



Diámetro del núcleo = 8-10μm (comercial)
Diámetro del recubrimiento = 250μm (comercial)
Contraste de índice Δn = 0.007 (*muy* baja NA)
Atenuación = 0.25dB/km

Reflexion interna total



Para ángulos mayores que el ángulo crítico, la onda se refleja CASI como si fuera un espejo.

En ambos casos hay una onda viajando cerca de la interfaz. Por esta razón toda reflexión interna total tiene una onda que decrece exponencialmente llamada: ONDA EVANESCENTE

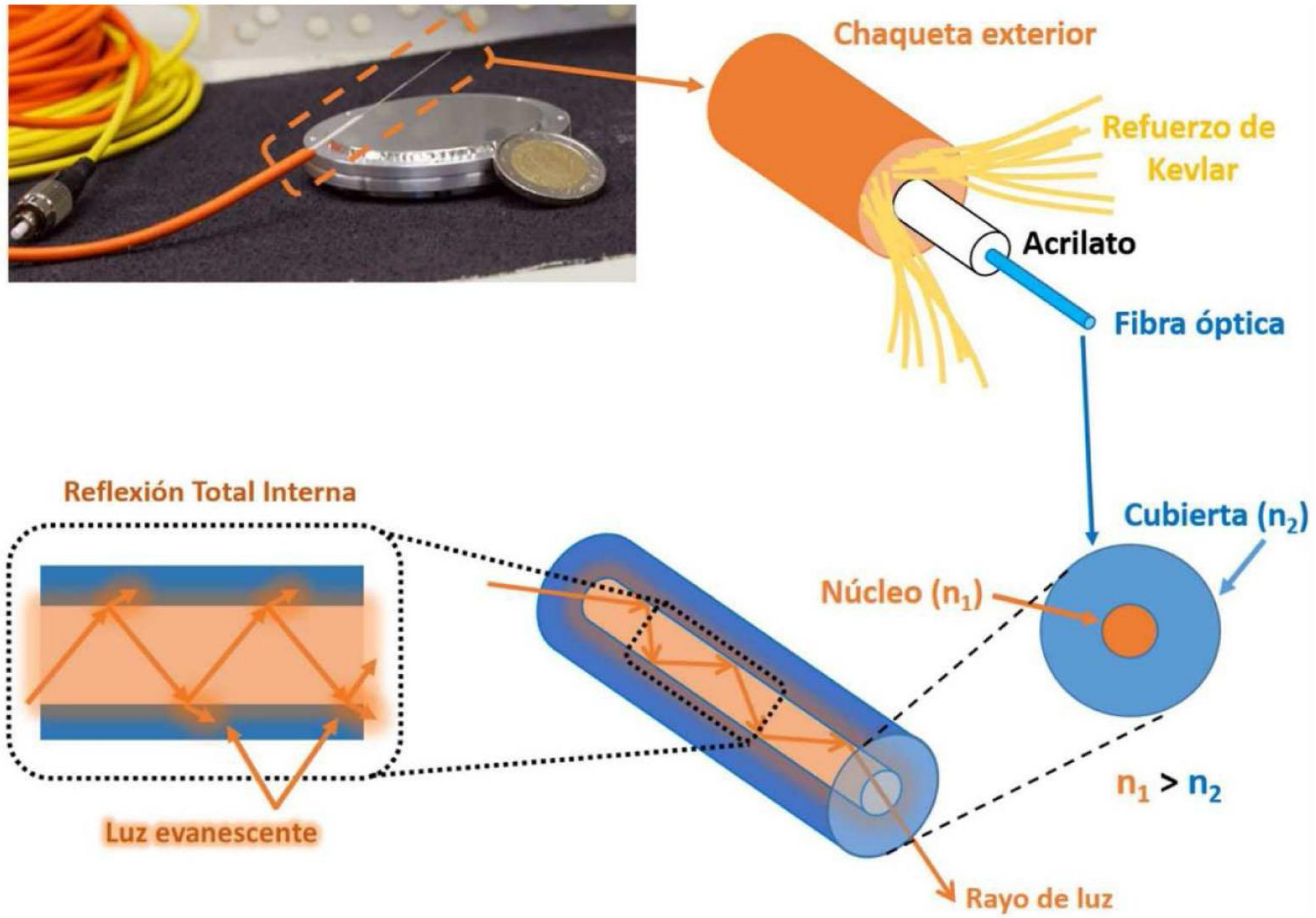
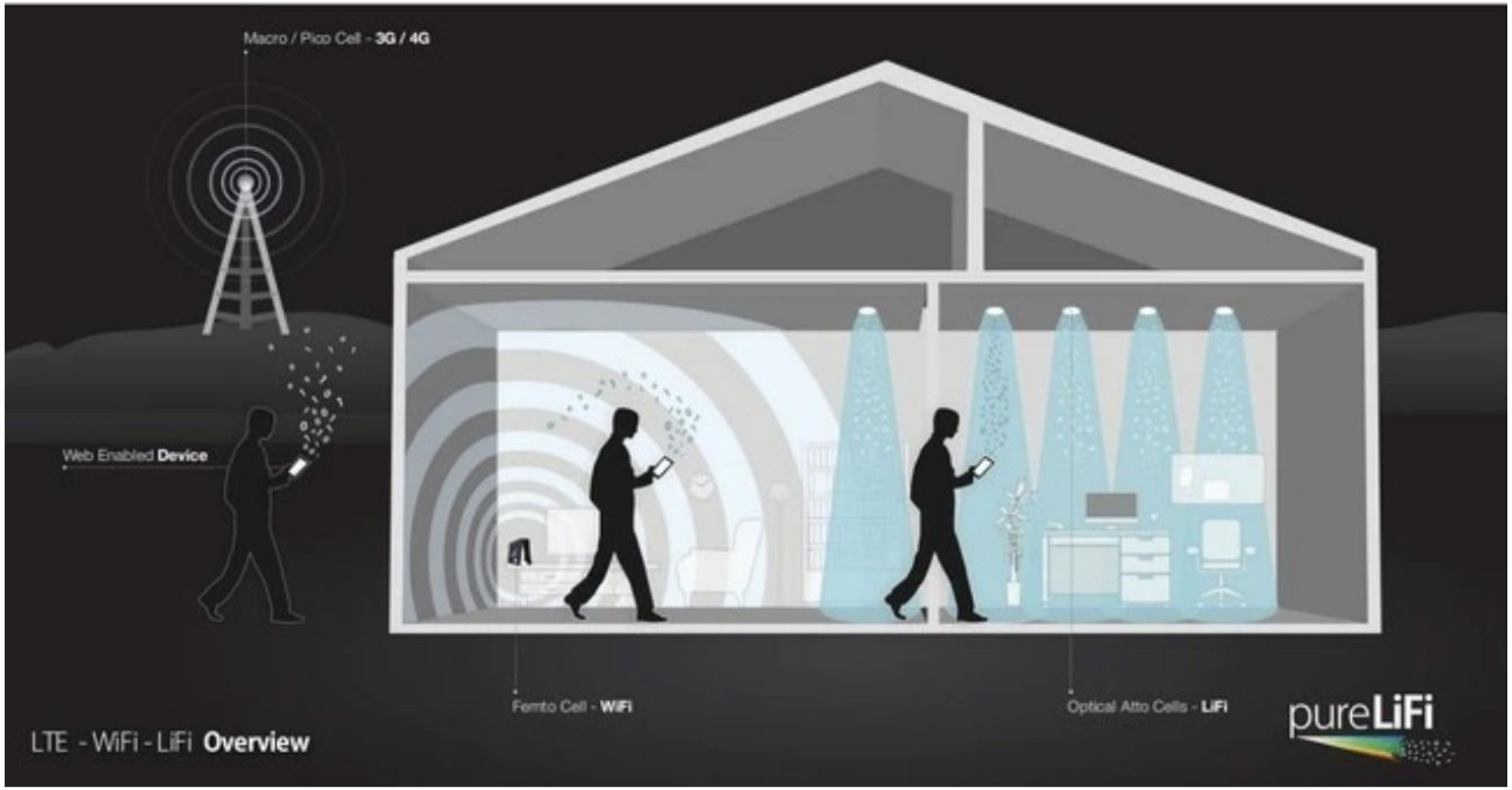


FIGURA 1. Propagación de la luz en el interior de un cable de fibra óptica.

A causa de la reflexión total interna, las fibras ópticas permiten que las señales viajen por largas distancias sin necesidad de repetidores (amplificadores), para compensar las reducciones en la intensidad de la señal.

Los repetidores de fibras ópticas, por lo general, están separados unos ~100 km, en comparación con una distancia de ~1.5 km que separa a los repetidores en los sistemas eléctricos (basados en cables).

Li-Fi

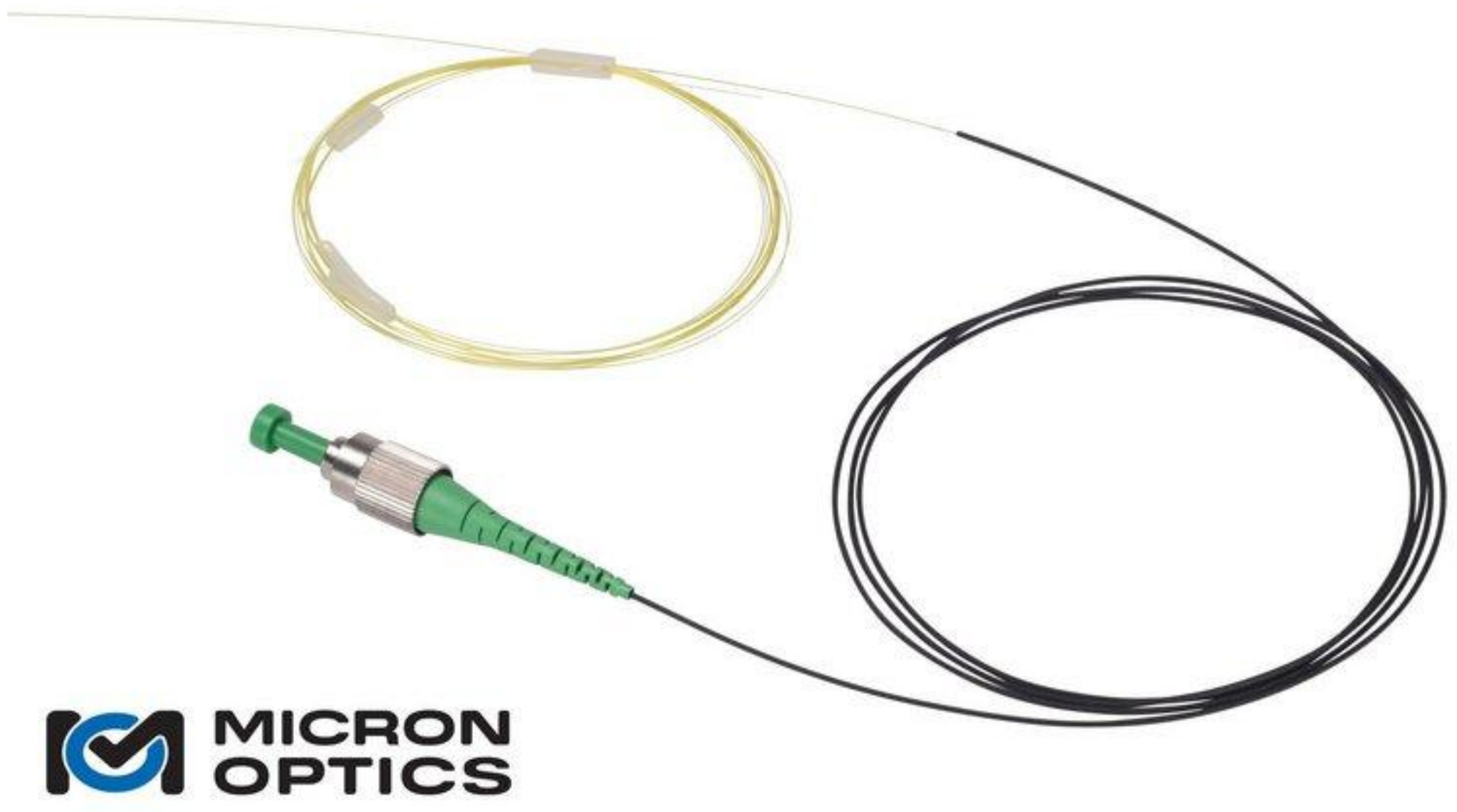


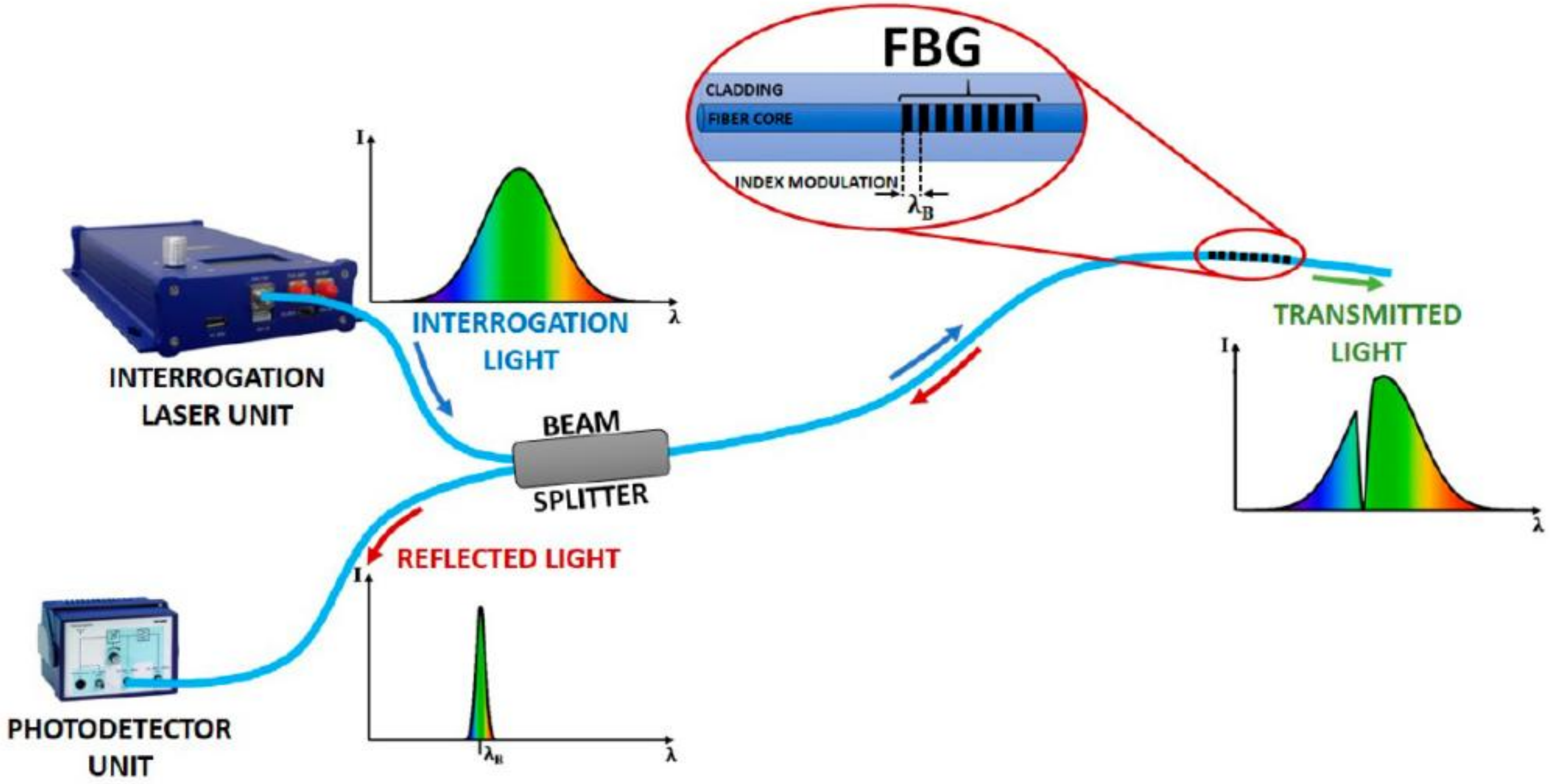
Perfecto para quirófanos o UCI donde no se permite Wi-Fi por interferencias electromagnéticas.

Uso de Li-Fi para conexión a internet en cabinas sin afectar los sistemas de navegación.

Transmisiones en espacios sensibles donde se requiere comunicación **segura, local y sin emisión externa**.

Fiber Branch Grating





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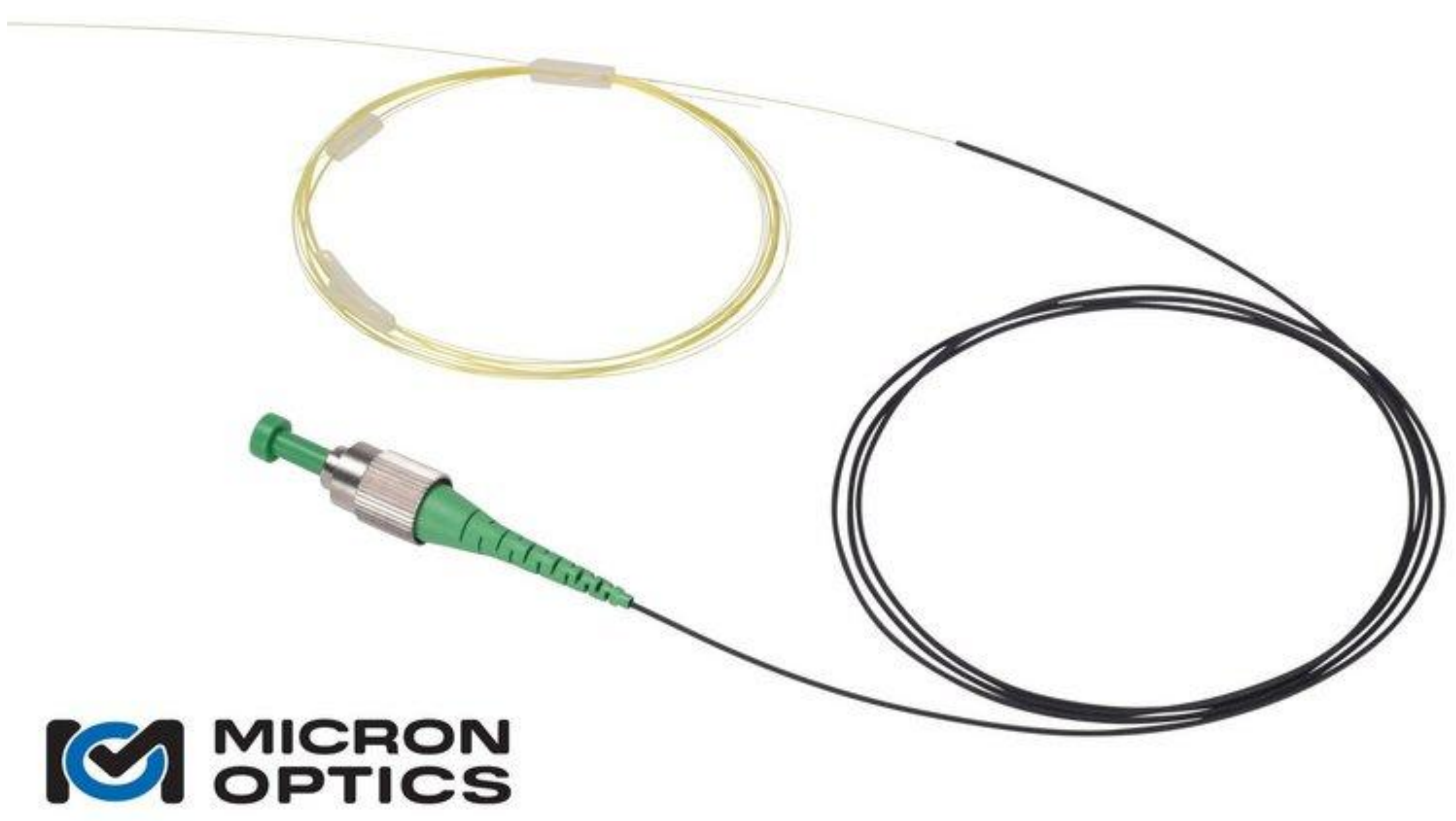
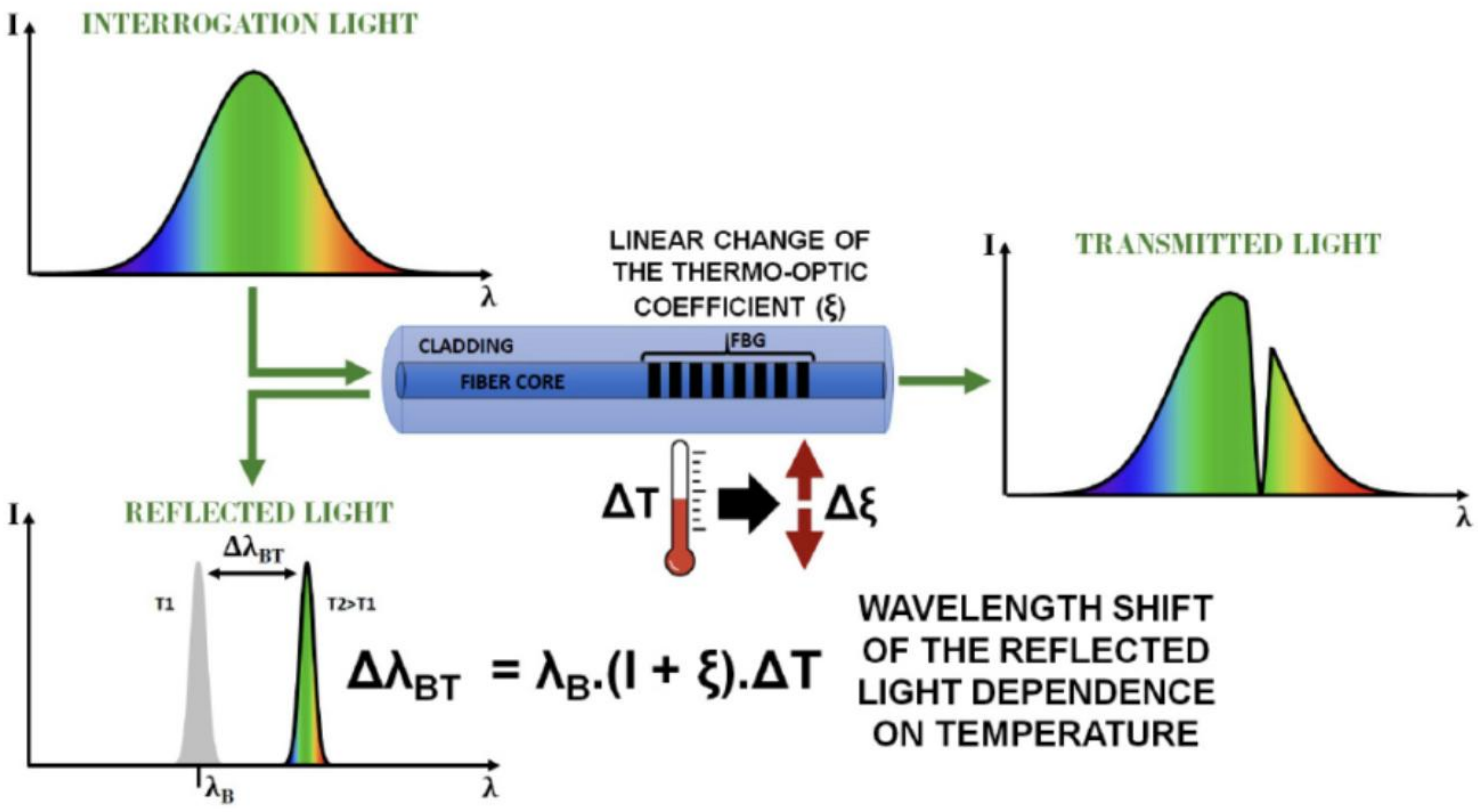
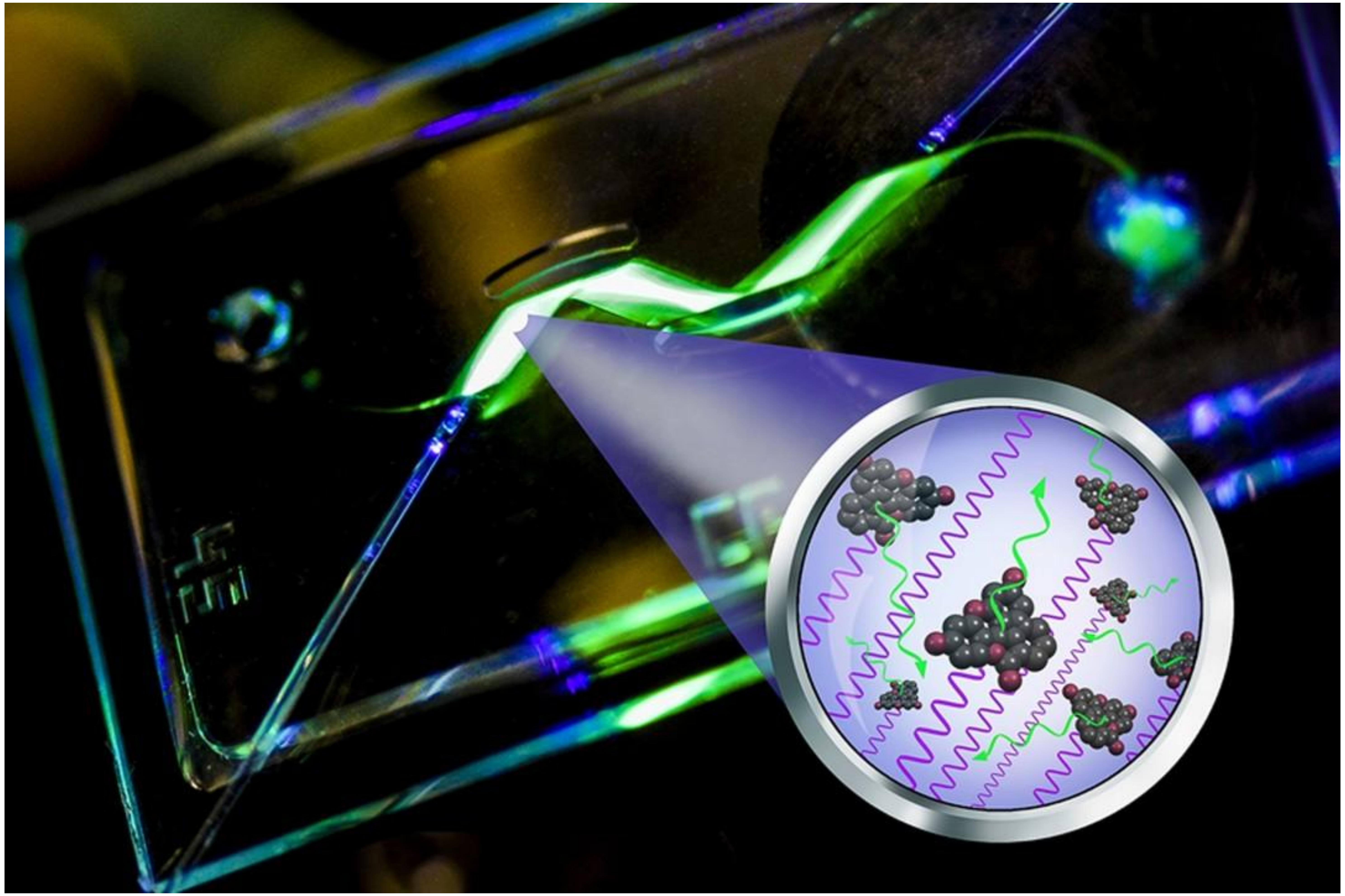
Fiber Branch Grating





Photonic Lab-on-a-Chip



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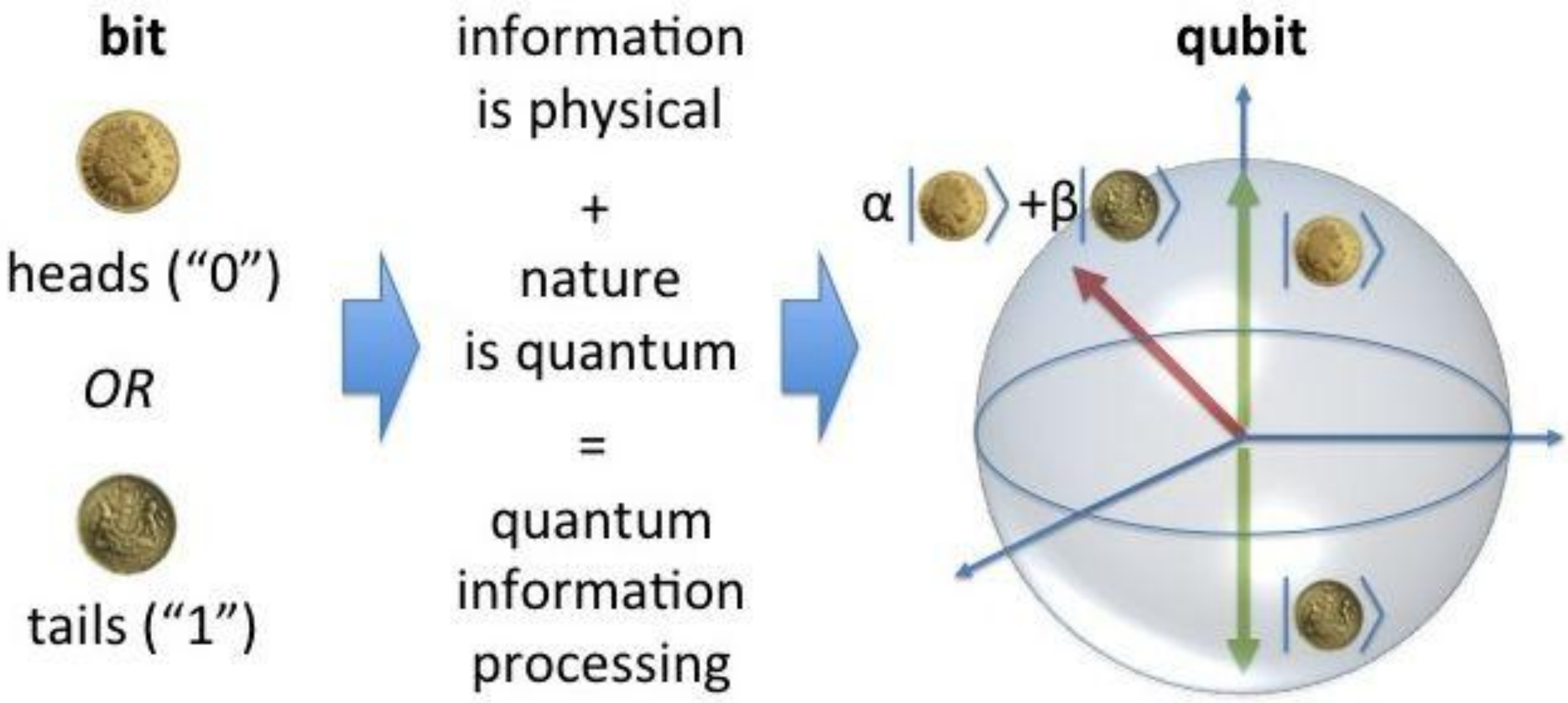
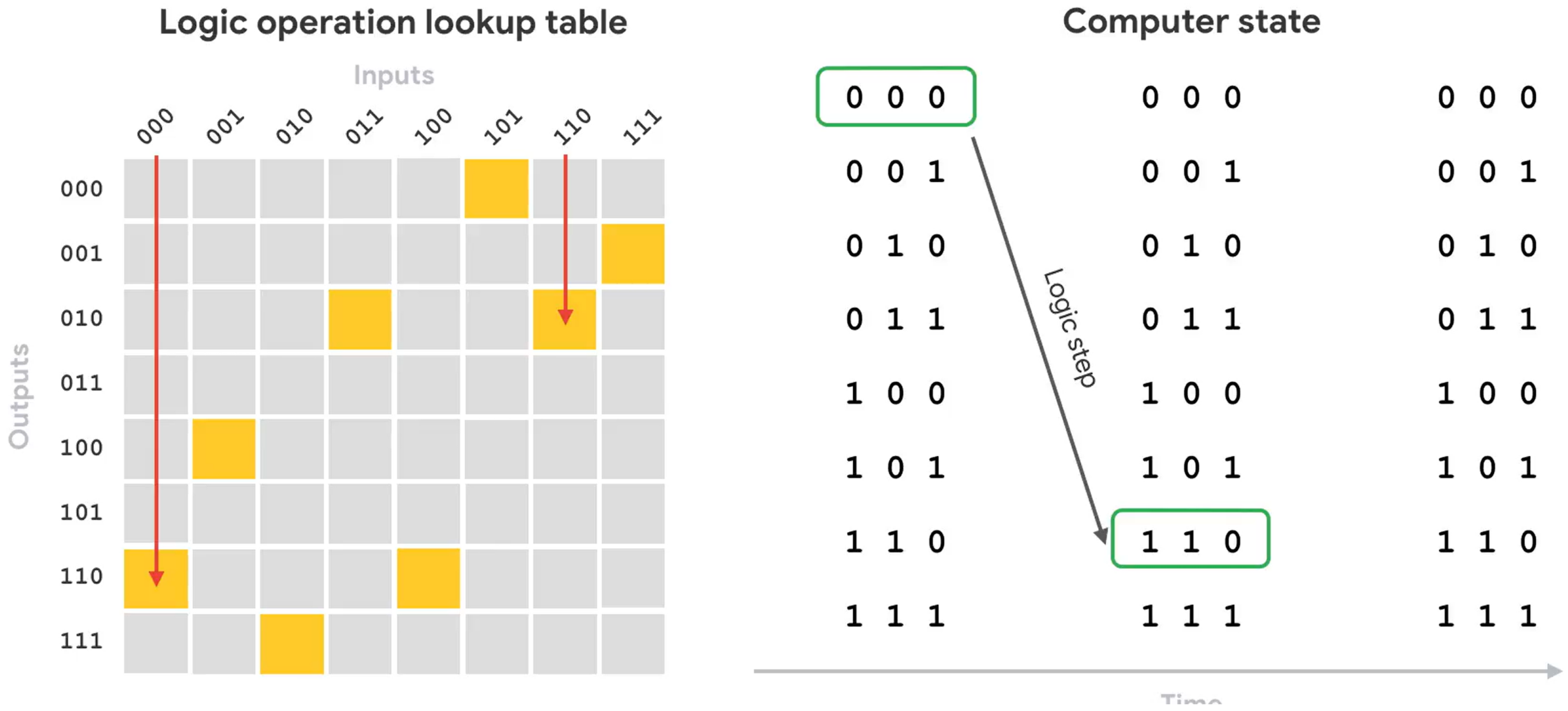
Outline

How to use quantum as information?

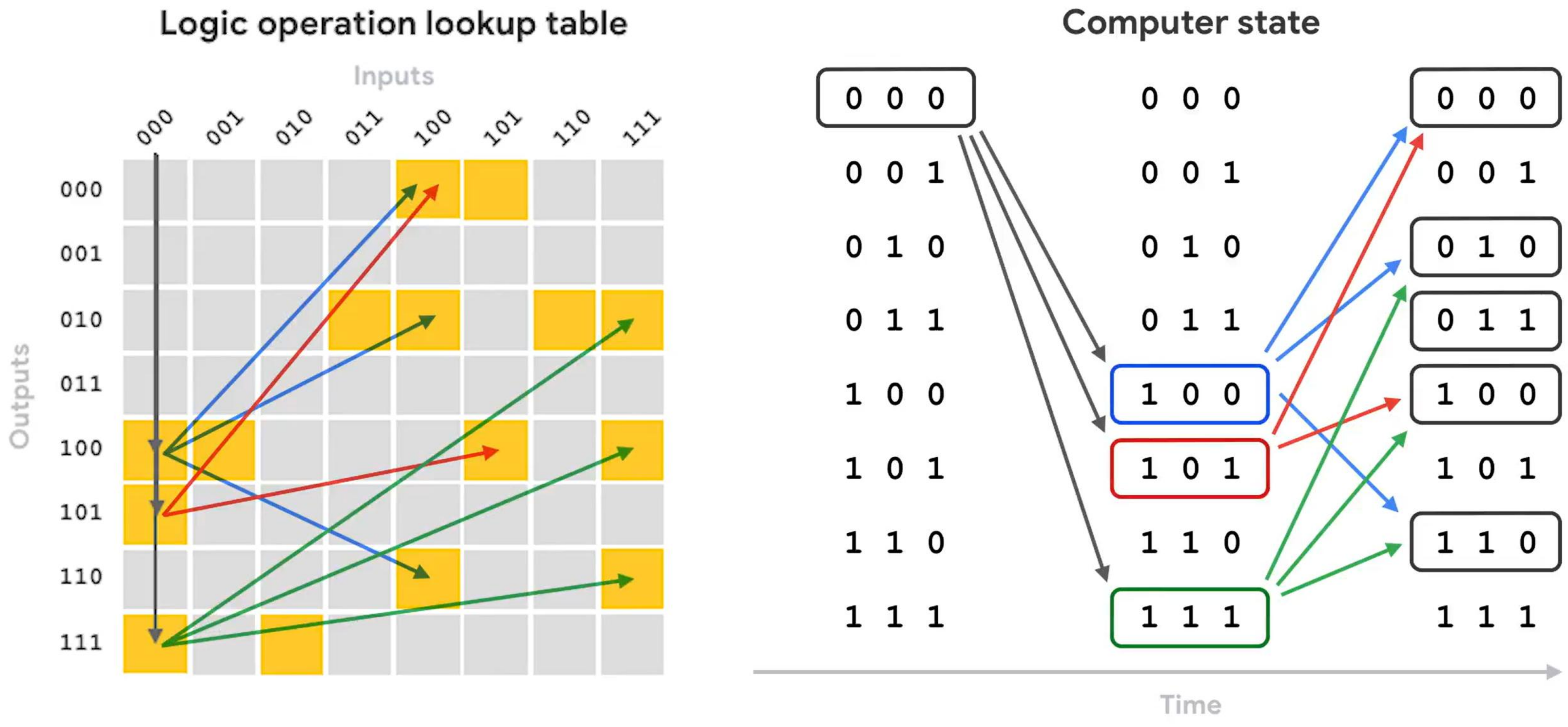


Logic operation states

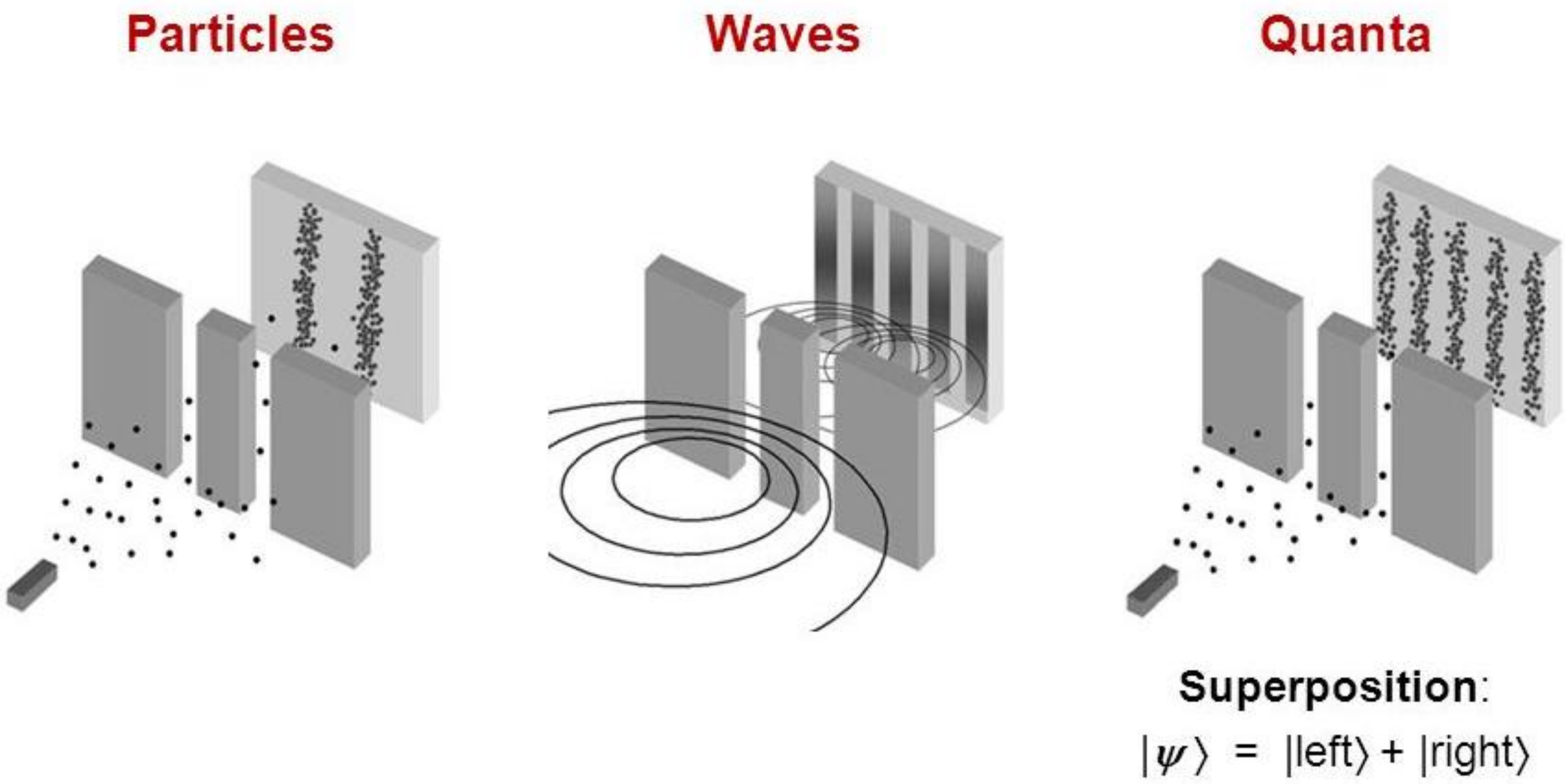
Classical



Quantum



Quantum measurement - Wave function collapse



Security

Post-Quantum Cryptography (PQC)

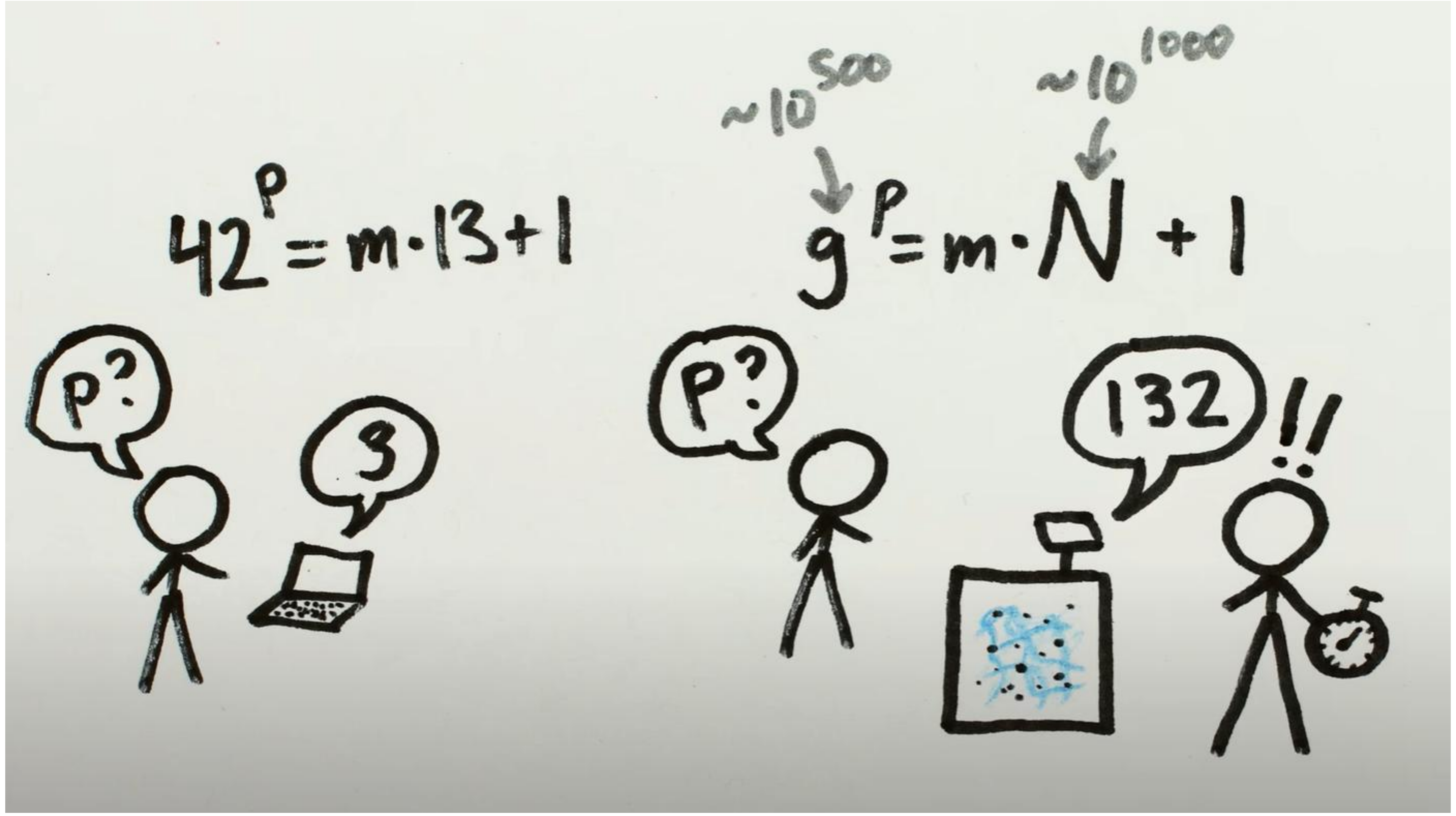
Es un conjunto de algoritmos criptográficos **clásicos** (funcionan en computadoras normales) que están diseñados para ser **seguros incluso frente a ataques con computadoras cuánticas**.

Quantum Key Distribution (QKD)

QKD es una técnica basada en **principios físicos cuánticos** (no matemáticos) para distribuir claves secretas entre dos partes con **seguridad garantizada por la física**.

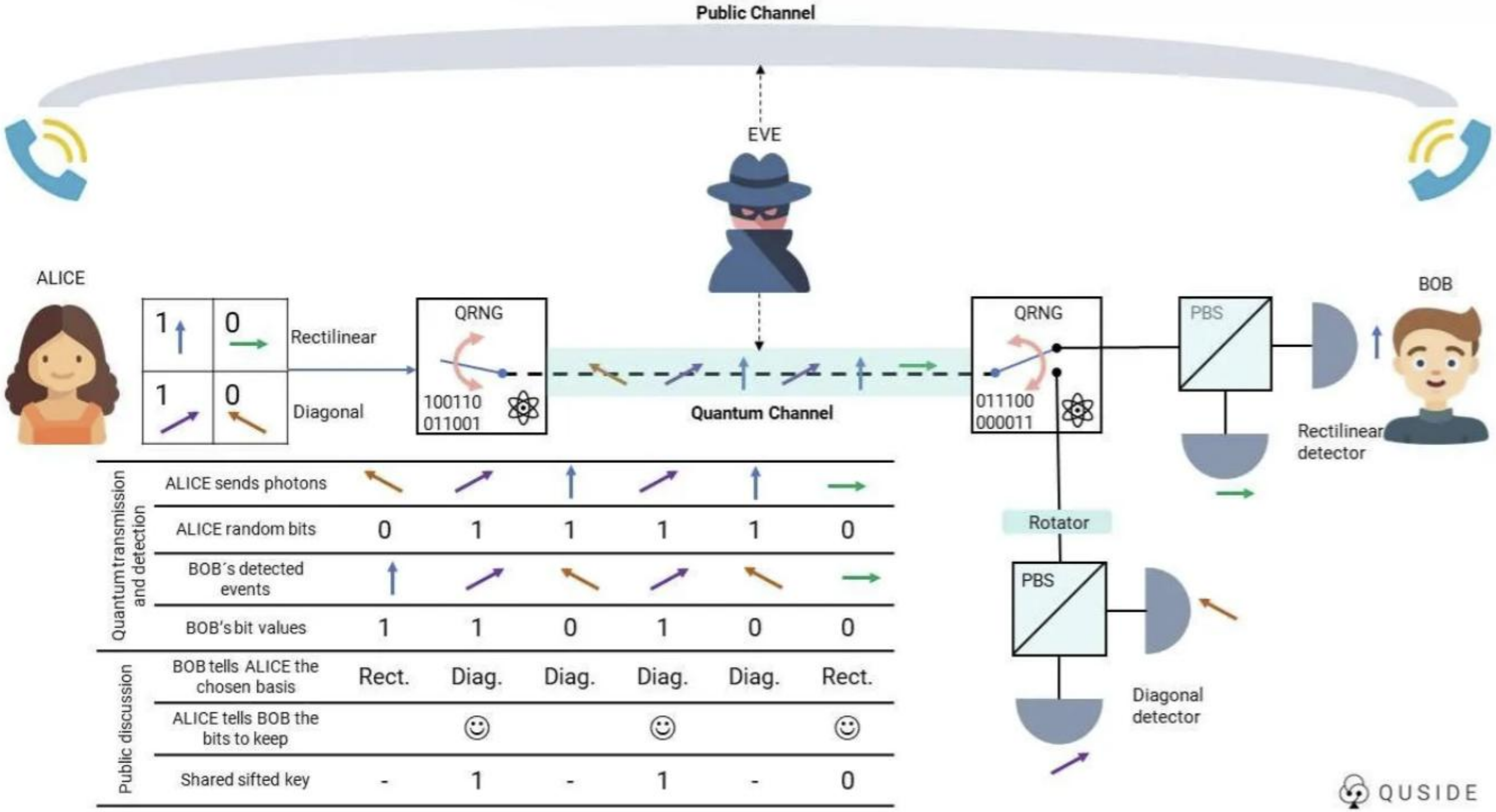
Post-Quantum Cryptography (PQC)

Shor's Algorithm



https://www.youtube.com/watch?v=lvTqbM5Dq4Q

Quantum Key Distribution (QKD)



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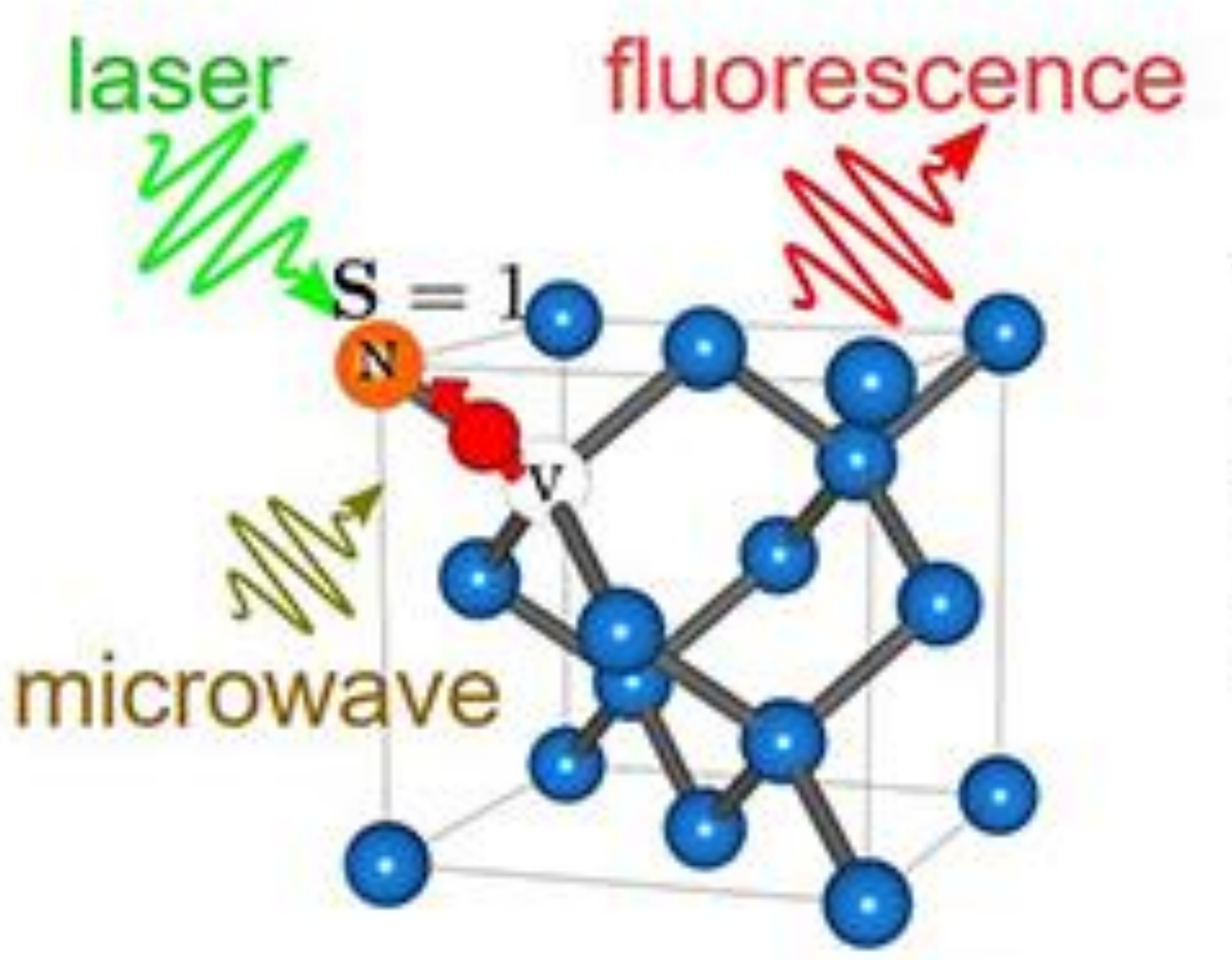
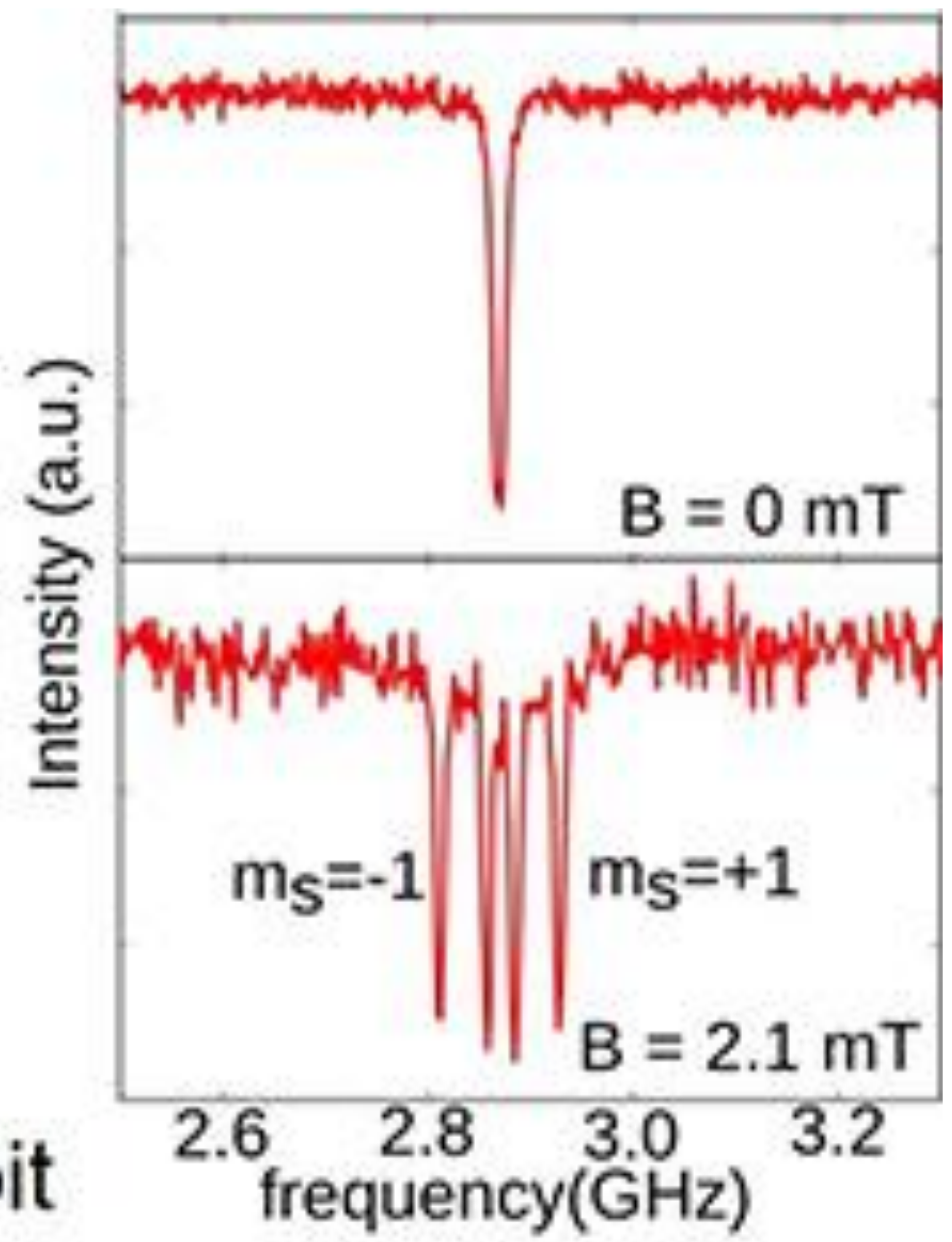
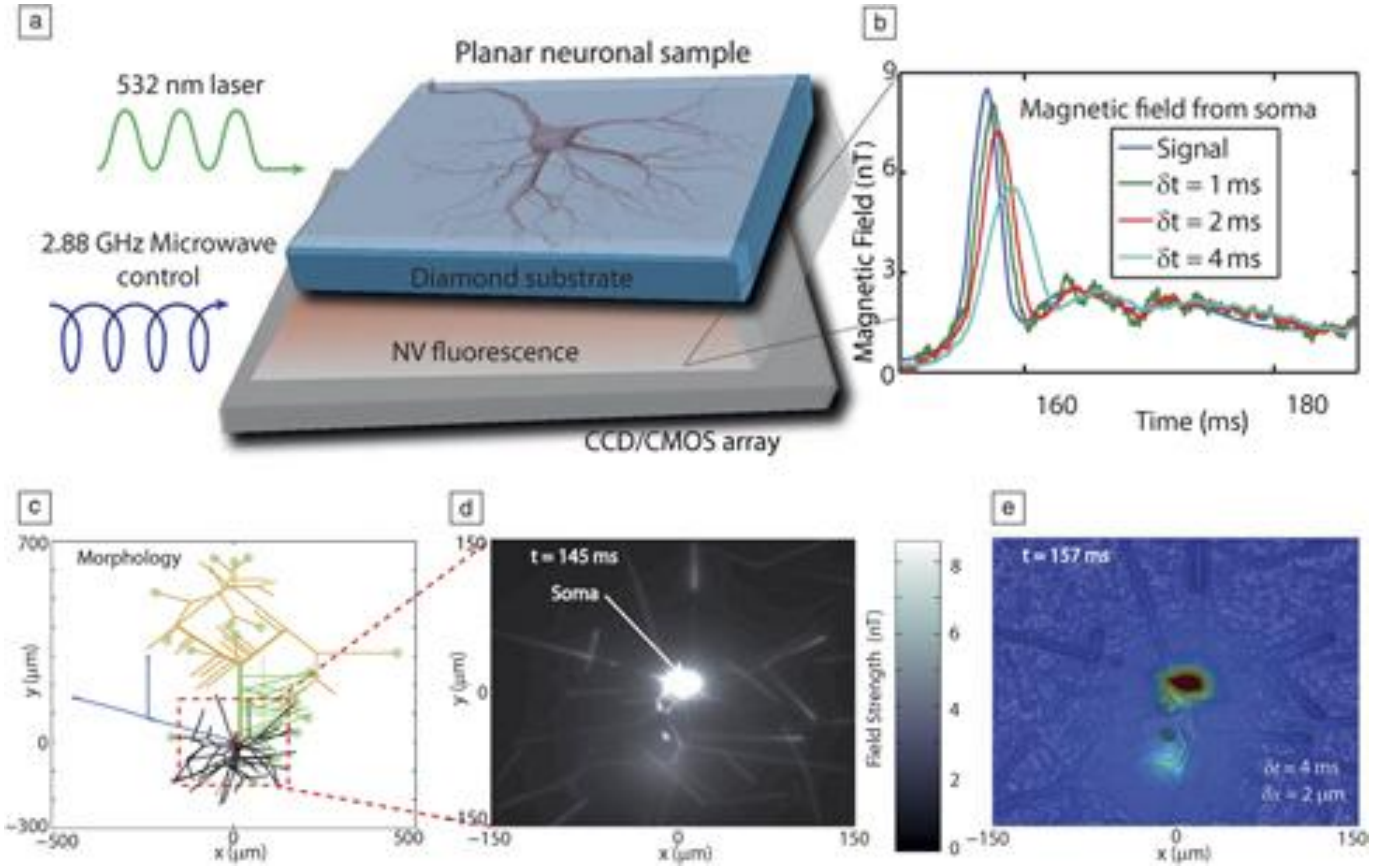
Nitrogen Vacancy Centers: Magnetic Field Sensors





- optically addressable qubit



Quantum internet?

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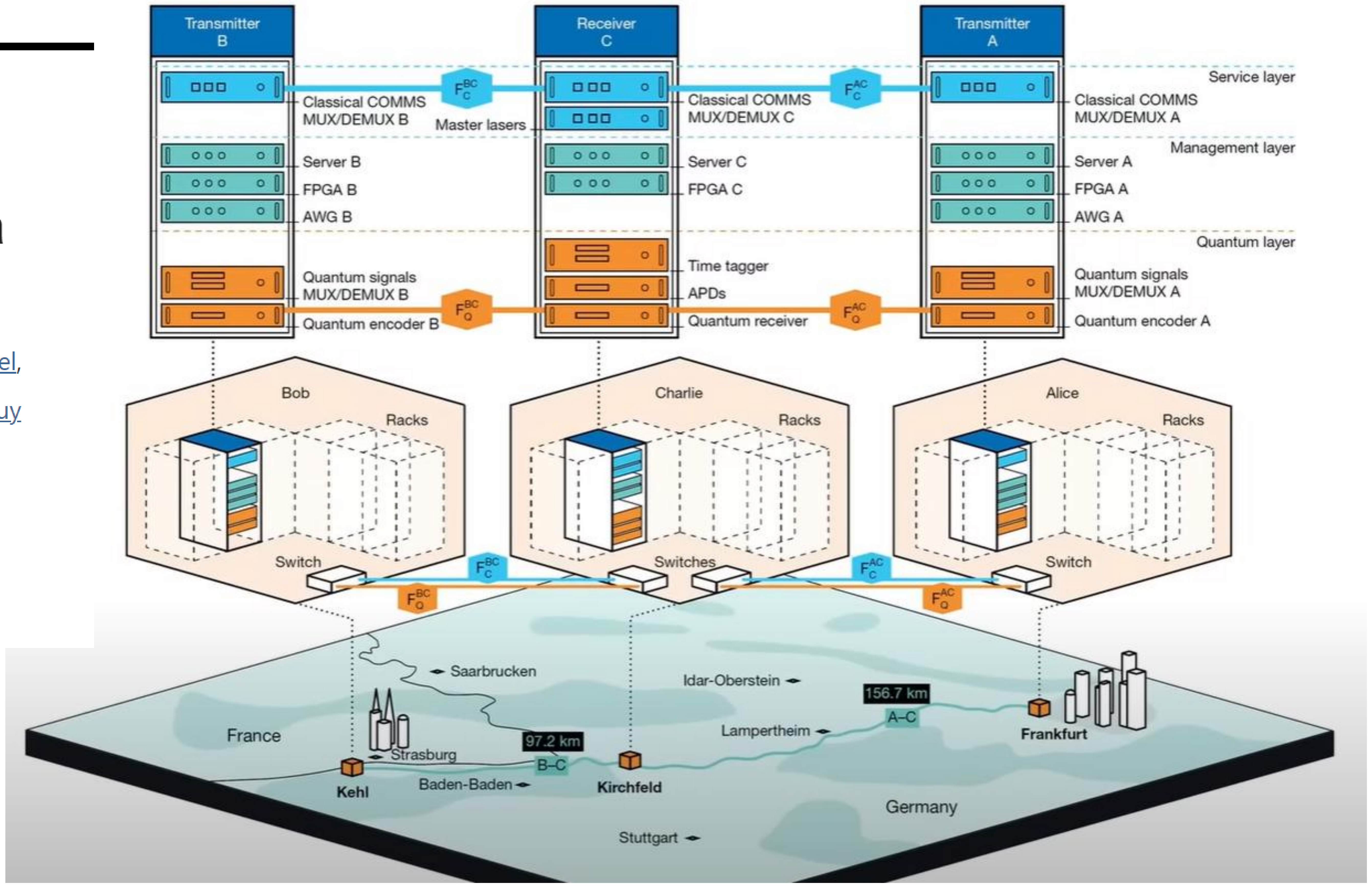
Article | Published: 23 April 2025

Long-distance coherent quantum communications in deployed telecom networks

Mirko Pittaluga, Yuen San Lo, Adam Brzosko, Robert I. Woodward, Davide Scalcon, Matthew S. Winnel, Thomas Roger, James F. Dynes, Kim A. Owen, Sergio Juárez, Piotr Rydlichowski, Domenico Vicinanza, Guy Roberts & Andrew J. Shields

Nature **640**, 911–917 (2025) | Cite this article



NSA/CSS

About Press Room Careers History

Quantum Key Distribution (QKD) and Quantum Cryptography (QC)

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Synopsis

NSA continues to evaluate the usage of cryptography solutions to secure the transmission of data in National Security Systems. NSA does not recommend the usage of quantum key distribution and quantum cryptography for securing the transmission of data in National Security Systems (NSS) unless the limitations below are overcome.

Outline

Introduction

Optics and photonics

Quantum technologies

Conclusion